Stochastic Approximation: From Robbins–Monro to Variance-Reduced Proximal Gradient Methods

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Acknowledgement of Sources

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Abstract

Many optimisation problems in statistics and machine learning have a finite-sum form $F(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$, where computing the full gradient is expensive for large datasets. Stochastic gradient descent (SGD) replaces ∇F by a cheap component gradient ∇f_i, but this introduces noise that typically does not disappear even when close to the minimiser. This report frames SGD and proximal SGD (Prox-SGD) as instances of Robbins–Monro stochastic approximation: the update consists of a stabilising mean field plus a martingale difference noise term. Under strong convexity, the drift contracts toward the unique minimiser, but persistent noise creates a steady-state error under constant step sizes, motivating classical diminishing step-size conditions.

Variance-reduced methods modify this noise model. For stochastic variance reduced gradient (SVRG) and proximal stochastic variance reduced gradient (Prox-SVRG), the mean field is unchanged, but the control-variate estimator yields a conditional variance bound that shrinks as both the iterate and the snapshot approach the solution, removing the variance floor that limits SGD. Controlled finite-sum quadratic experiments on strongly convex ridge least squares and a composite ridge+ℓ_1 objective support this interpretation: constant-step SGD and Prox-SGD exhibit a plateau, Robbins–Monro step-size schedules improve accuracy slowly, and (Prox-)SVRG exhibits near-geometric decay with estimator variance collapsing toward numerical precision.

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Introduction

Many optimisation problems in statistics and machine learning can be written as finite-sum objectives $F(x) = \frac{1}{n}\sum_i f_i(x)$, where each f_i corresponds to the contribution of one data point. In large-scale settings the full gradient $\nabla F(x)$ is expensive to compute, so stochastic methods replace it with a cheaper random component gradient $\nabla f_i(x)$. This simple idea leads to stochastic gradient descent (SGD), which is still the default baseline algorithm in modern applications.

Stochastic Approximation (SA). The drawback is that stochastic gradients are noisy. Even when the iterate is close to the minimiser, the update direction typically remains random, limiting how far SGD can go with a constant step size. Classical stochastic approximation (Classical SA), beginning with Robbins–Monro, provides a clean way to understand this behaviour. In the Classical SA recursion, the update is driven by a deterministic drift term (the mean field) plus a random noise term. Under strong convexity the drift is stabilising and pulls the iterates toward the unique minimiser, but persistent noise creates a steady-state error unless step sizes decay.

Variance-reduced methods were designed specifically to address this bottleneck. Algorithms such as stochastic variance reduced gradient (SVRG) (and its proximal variant Prox-SVRG for composite objectives) still use stochastic component gradients, but they build control-variate estimators whose variance shrinks as the method approaches the solution. Empirically, this often changes the whole "shape" of convergence curves: instead of plateauing with a constant step size, variance-reduced methods can continue improving at a geometric rate in strongly convex problems.

The goal of this report is to connect these observations to the classical SA viewpoint and to make the change in the noise model explicit. In this paper, we explore how the Robbins–Monro stochastic approximations explain the behaviour of SGD and proximal SGD, and what precisely changes in the noise process when moving to variance-reduced methods such as (Prox-)SVRG, so that constant step sizes can achieve linear convergence under strong convexity.

Chapter 1 reviews the relevant background on stochastic approximation, stochastic gradient, proximal methods, and variance reduction techniques. Chapter 2 bridges Robbins–Monro theory with modern optimisation, showing how SGD and Prox-SGD fit into the stochastic approximation recursion and explaining why persistent noise requires diminishing step sizes. Chapter 3 develops the stochastic-approximation interpretation of SVRG and Prox-SVRG and derives a variance bound that shrinks near the solution, enabling linear convergence under strong convexity. Chapter 4 presents controlled finite-sum quadratic experiments and connects the empirical results back to the stochastic-approximation predictions. Chapter 5 summarises the conclusions and highlights limitations and possible extensions.

Chapter 1

Background and Related Work

Classical Stochastic Approximation (Classical SA) studies iterative procedures for solving equations or optimisation problems when only noisy information is available. The central idea is to replace an intractable deterministic computation by a sequence of cheap random measurements, and to average out the noise through a carefully chosen step-size schedule.

The Robbins–Monro root finding framework [1] The modern stochastic approximation (Modern SA) literature begins with Robbins–Monro, which posed a canonical problem: find a root x^* of an unknown function h, where $h(x)$ cannot be evaluated directly.

Instead, at iteration k we observe a noisy measurement $Y_{k+1}(x)$ (e.g. from simulation, random experiments, or sampling). The Robbins–Monro update takes the form

$$x_{k+1} = x_k - \alpha_k Y_{k+1}(x_k) \qquad \text{where } (\alpha_k)_{k\geq 0} \text{ is a step-size sequence.} \tag{1.0.1}$$

Writing the observation as $Y_{k+1}(x_k) = h(x_k) + \varepsilon_{k+1}$ with $\mathbb{E}[\varepsilon_{k+1} \mid \mathcal{F}_k] = 0$ makes the core SA structure explicit

$$x_{k+1} = x_k - \alpha_k\big(h(x_k) + \varepsilon_{k+1}\big). \tag{1.0.2}$$

This is the conceptual template of SA: the drift term $h(x_k)$ determines stability, whilst the noise term ε_{k+1} determines how aggressively one can step.

Step sizes, drift stability, and the noise barrier The Robbins–Monro Framework shows that convergence is driven by a balance between (i) making enough cumulative progress toward x^* and (ii) preventing random fluctuations from accumulating indefinitely. This balance is encoded in the classical step-size conditions

$$\sum_{k=0}^{\infty} \alpha_k = \infty, \qquad \text{and} \qquad \sum_{k=0}^{\infty} \alpha_k^2 < \infty, \tag{1.0.3}$$

where the first condition ensures that the algorithm does not stall too early, while the second ensures that the total noise from the updates stays finite.

Step-size control alone is not sufficient. The drift itself must be stabilising (the drift itself must naturally pull you towards the solution). A convenient sufficient condition for stability is

the strong monotonicity inequality

$$\langle h(x), x - x^* \rangle \geq \mu \|x - x^*\|^2, \qquad \text{for some } \mu > 0.$$

This says that, away from the root, the drift points back toward x^* with strength proportional to the squared distance. Under this condition, the drift acts like a restoring force, while the noise acts like continual random deviations.

This perspective highlights an issue that shows up repeatedly in stochastic optimisation: if the noise does not die out near x^*, then constant step sizes typically produce a non-zero error floor. Diminishing step sizes are therefore not a technical detail, but the mechanism that makes the accumulated effect of noise controllable.

Recursion to analysis [1] To analyse (1.0.2) properly, the literature defines unbiased noise in the following way: write $\mathcal{F}_k$ for "everything revealed up to iteration k" (past iterates and past randomness), and then assume the next noise term has zero conditional mean

$$\mathbb{E}[\varepsilon_{k+1} \mid \mathcal{F}_k] = 0.$$

This is a formal way of saying the new observation does not introduce a systematic bias: given the past, the average update direction is still the drift $h(x_k)$ (e.g., standard Robbins–Monro expositions). Once this conditional-mean-zero structure is in place, many of the convergence proofs boil down to: each step makes progress because the drift pushes toward x^*, but that progress is partially disturbed by random noise. The filtration matters because it makes this split precise: at iteration k, x_k is already determined by the past (so it is "fixed" when conditioning on $\mathcal{F}_k$), and the only new randomness comes from the next sample used to form the update at step $k + 1$.

1.1 Stochastic Gradient Methods

Stochastic approximation is a central tool in modern optimisation because many learning problems naturally have a finite-sum structure. In empirical risk minimisation (ERM), we fit parameters $x \in \mathbb{R}^d$ by minimising an average loss over a dataset $\{z_i\}_{i=1}^n$,

$$F(x) = \frac{1}{n} \sum_{i=1}^{n} f_i(x), \qquad f_i(x) = \ell(x; z_i), \tag{1.1.1}$$

often with additional regularisation. In large-scale problems, computing $\nabla F(x)$ exactly costs n component gradients per iteration, so first-order methods are typically built from cheap random estimates based on one (or a small batch of) data point(s) at a time [2, 3].

SGD for finite-sum problems Stochastic gradient descent (SGD) replaces the full gradient $\nabla F(x_k)$ by a randomly sampled component gradient. With $i_{k+1} \sim \mathrm{Unif}\{1, \ldots, n\}$, the update is

$$x_{k+1} = x_k - \alpha_k \nabla f_{i_{k+1}}(x_k). \tag{1.1.2}$$

The main property that makes this work is unbiasedness: conditioning on the information available at step k fixes x_k, and uniform sampling gives

$$\mathbb{E}[\nabla f_{i_{k+1}}(x_k) \mid \mathcal{F}_k] = \frac{1}{n}\sum_{i=1}^{n} \nabla f_i(x_k) = \nabla F(x_k),$$

so the stochastic gradient is an unbiased estimator of the true gradient [2, 3].

It is often helpful to express (1.1.2) in the same "drift + noise" terms as classical SA by writing

$$\nabla f_{i_{k+1}}(x_k) = \nabla F(x_k) + \xi_{k+1}, \qquad \xi_{k+1} := \nabla f_{i_{k+1}}(x_k) - \nabla F(x_k),$$

so that $\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = 0$. In this form, the drift is the true gradient $\nabla F(x_k)$ and the noise term, ξ_{k+1}, captures sampling variability [2].

The persistent-noise barrier and step-size implications A main theme in the SGD literature is that unbiasedness alone does not guarantee that the noise becomes small near the solution. Even if F is strongly convex with minimiser x^* and $\nabla F(x^*) = 0$, the component gradients typically do *not* vanish at x^*

$$\xi_{k+1}(x^*) = \nabla f_{i_{k+1}}(x^*) - \nabla F(x^*) = \nabla f_{i_{k+1}}(x^*),$$

so the stochastic gradient remains random unless all samples share the same stationary point. This persistent variance at the optimum creates a fundamental bottleneck. As detailed by Bottou et al. [4], a constant step size α restricts SGD to a neighbourhood of the optimum whose radius scales with $\mathcal{O}(\alpha)$. Escaping this noise floor requires enforcing the classical Robbins–Monro condition $\sum \alpha_k^2 < \infty$, but dampening the noise in this way limits the algorithm to a sublinear convergence rate [2–4].

In strongly convex smooth settings, the literature therefore distinguishes two practical methods. With constant step sizes, SGD rapidly approaches a neighbourhood of x^* whose radius is controlled by the step size and the noise level. With diminishing step sizes (such as Robbins–Monro schedules), the method can converge more accurately, but the tail behaviour becomes sublinear because steps eventually become very small [2, 3]. Framing the issue around this persistent-noise viewpoint helps to clarify the motivation for variance reduced methods [5].

1.2 Proximal Stochastic Methods

Many optimisation problems in statistics and machine learning are naturally composite

$$P(x) = F(x) + R(x), \tag{1.2.1}$$

where F is smooth (often a finite-sum empirical risk) and R is convex but possibly non-smooth. Typical choices of R include ℓ_1 penalties for sparsity and indicator functions for constraints. In these settings, a plain gradient step on P is not available because ∇R may not exist, so first-order methods replace explicit differentiation of R by a proximal step [3, 5].

The proximal operator and composite first-order conditions [5]

Definition 1.2.1 (Proximal Operator)**.** *For a proper, lower semicontinuous, convex function R, the proximal operator is defined by*

$$\operatorname{prox}_{\alpha R}(x) = \arg\min_{z\in\mathbb{R}^d} \left\{ R(z) + \frac{1}{2\alpha}\|z-x\|^2 \right\}, \qquad \alpha > 0.$$

The proximal operator encodes the non-smooth structure through an implicit optimisation step rather than a gradient evaluation. A standard result of this definition is the optimality condition

$$\frac{1}{\alpha}\big(x - \operatorname{prox}_{\alpha R}(x)\big) \in \partial R(\operatorname{prox}_{\alpha R}(x))\,.$$

When F is differentiable, the deterministic proximal-gradient method applies a gradient step to F followed by a proximal step for R,

$$x_{k+1} = \operatorname{prox}_{\alpha R}(x_k - \alpha\nabla F(x_k))\,. \tag{1.2.2}$$

This update corresponds to the composite first-order condition $0 \in \nabla F(x^*) + \partial R(x^*)$, and can be viewed as a forward (gradient) step on F coupled with a backward (proximal) step on R [5].

Prox-SGD In the finite-sum setting, proximal stochastic gradient methods replace $\nabla F(x_k)$ in (1.2.2) by a component gradient $\nabla f_{i_{k+1}}(x_k)$, giving the Prox-SGD update

$$x_{k+1} = \operatorname{prox}_{\alpha_k R}\big(x_k - \alpha_k \nabla f_{i_{k+1}}(x_k)\big)\,, \qquad i_{k+1} \sim \operatorname{Unif}\{1,\dots,n\}. \tag{1.2.3}$$

As with SGD, the stochasticity comes from sampling, while the proximal operator handles the non-smooth component in a principled way [2, 5]. From a noise perspective, the proximal step does not remove the sampling variability in the gradient estimate; it only changes how the non-smooth part is treated. This results in the same issue seen in classical SGD. Unless the underlying stochastic gradient noise vanishes at the solution, constant step sizes lead to a steady-state error, while diminishing step sizes improve accuracy at the cost of slower late-stage progress [2, 3]. Hence, proximal variance-reduced methods are of interest in the finite-sum composite setting: they keep the proximal structure needed for R while modifying the gradient estimator, so that its variance shrinks as the iterates approach a minimiser [5, 6].

1.3 Variance Reduction Techniques (SVRG)

Unbiasedness alone is not enough for fast convergence. Even if $\mathbb{E}[g(x)] = \nabla F(x)$, the variance of $g(x)$ can remain non-zero near a minimiser. In strongly convex problems this usually creates a "variance floor": with a constant step size, the iterates keep fluctuating in a neighbourhood of x^* rather than settling at it, while diminishing step sizes trade accuracy for slower late-stage progress [2, 3]. Variance reduction methods are designed to change this noise behaviour by constructing gradient estimators whose variance shrinks as the algorithm approaches the solution, without losing the cheap per-iteration cost of component gradients.

Variance-reduction paradigms The finite-sum structure $F(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$ makes it possible to reduce variance in two broad ways.

1. Incremental aggregated-gradient methods (SAG/SAGA). These methods maintain a memory of past component gradients and update an aggregated estimate at each iteration. An example of this is SAG, which achieves fast (often linear) convergence behaviour on finite training sets by continuously refining a stored gradient table rather than repeatedly sampling fresh "from scratch" gradients [7]. The key idea is that as more components are kept up to date, the aggregated direction becomes less random and more like the full gradient.

2. Control-variate methods (SVRG family). This family of algorithms builds an explicit control variate by anchoring gradients at a reference point ("snapshot") and correcting fresh samples using component-wise differences. This family includes SVRG and its proximal extensions [5, 6]. At a high level, these methods keep the same mean direction as SGD (the full gradient), but modify the noise so that it is tied to how far the current iterate is from the snapshot.

Stochastic Variance Reduced Gradient (SVRG) [6] alternates between (i) computing a full gradient at a reference point $\tilde{x}$ and (ii) performing many cheap stochastic updates using a corrected estimator. With $i \sim \mathrm{Unif}\{1, \ldots, n\}$, the estimator is

$$g(x; \tilde{x}, i) = \nabla f_i(x) - \nabla f_i(\tilde{x}) + \nabla F(\tilde{x}). \tag{1.3.1}$$

The associated update is $x^+ = x - \alpha\, g(x; \tilde{x}, i)$ with a typically constant step size α [6]. Two properties distinguish (1.3.1) from the plain stochastic gradient $\nabla f_i(x)$. *First*, it is unbiased, i.e.,

$$\mathbb{E}[g(x; \tilde{x}, i)] = \nabla F(x)$$

under uniform sampling. *Second*, its variability is driven by a *difference of gradients* evaluated at x and $\tilde{x}$, rather than by $\nabla f_i(x)$ alone. Under standard smoothness assumptions, if each f_i is L-smooth then

$$||\nabla f_i(x) - \nabla f_i(\tilde{x})|| \leq L||x - \tilde{x}||.$$

This implies that the conditional variance of the estimator is bounded by a constant multiple of $||x - \tilde{x}||^2$. Hence the variance contracts as the iterates remain close to the snapshot.

In particular, as both x and the snapshot approach a minimiser, the correction term shrinks and the estimator becomes increasingly deterministic, removing the fixed variance floor that limits constant-step SGD [2, 6]. This "vanishing-noise" mechanism explains why SVRG-type methods can achieve linear convergence rates on strongly convex finite-sum objectives with constant step sizes.

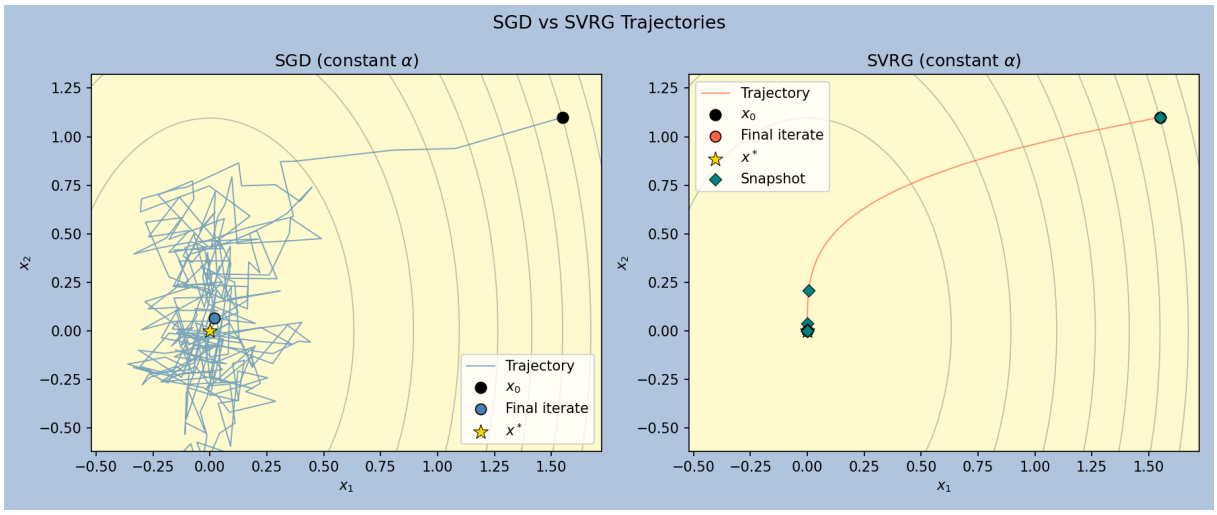

Figure 1.1: Shows illustrative 2D trajectories near x^*. With a constant step size, SGD exhibits a noise floor and fluctuates in a neighbourhood of x^*. In contrast, SVRG contracts tightly toward x^* as the control-variate estimator reduces variance near the solution. This illustrates the noise-reduction mechanism predicted by the stochastic approximation interpretation.

Variance reduction for composite objectives Given a composite objective $P(x) = F(x) + R(x)$, where R is convex and possibly non-smooth. Prox-SVRG extends SVRG by keeping the same variance-reduced estimator (1.3.1) for the smooth part F, while incorporating R through a proximal step:

$$x^+ = \mathrm{prox}_{\alpha R}\big(x - \alpha\, g(x; \tilde{x}, i)\big). \tag{1.3.2}$$

Xiao and Zhang [5] show that this hybrid preserves the core control-variate mechanism while remaining compatible with non-smooth regularisers such as ℓ_1 penalties and constraints [5]. Put simply, proximality addresses non-smooth structure, while variance reduction addresses the sampling-noise. The two components target different problems.

1.4 Mathematical Preliminaries

We operate on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_k)_{k\geq 0}, \mathbb{P})$, where Ω denotes the sample space of all complete algorithmic trajectories, equipped with the σ-algebra of measurable events $\mathcal{F}$ and probability measure $\mathbb{P}$.

The filtration $(\mathcal{F}_k)_{k\geq 0}$ represents the information generated by the algorithm up to iteration k. The iterate x_k is assumed to be $\mathcal{F}_k$-measurable, and the noise sequence $(\xi_{k+1})_{k\geq 0}$ satisfies the martingale difference property $\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = 0$ almost surely. This condition formalises the unbiasedness of the stochastic updates with respect to the information available at iteration k.

We consider stochastic iterative algorithms defined by the update rule:

$$x_{k+1} = x_k - \alpha_k\big(h(x_k) + \xi_{k+1}\big), \qquad k \geq 0, \tag{1.4.1}$$

where $(\alpha_k)_{k\geq 0}$ is a sequence of positive step sizes, $h : \mathbb{R}^d \to \mathbb{R}^d$ is a deterministic mean field, and $(\xi_{k+1})_{k\geq 0}$ is a noise sequence adapted to the filtration $(\mathcal{F}_k)$. The step-size sequence $(\alpha_k)_{k\geq 0}$ is

assumed to satisfy the classical Robbins–Monro conditions

$$\sum_{k=0}^{\infty} \alpha_k = \infty, \qquad \sum_{k=0}^{\infty} \alpha_k^2 < \infty. \tag{1.4.2}$$

Additionally, we assume that the noise sequence satisfies a bounded second-moment condition:

$$\mathbb{E}[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k] \leq \sigma^2 \qquad \text{almost surely, for some } \sigma^2 > 0.$$

We further assume that h is Lipschitz continuous and admits a unique root x^* satisfying

$$\langle h(x), x - x^* \rangle \geq \mu \|x - x^*\|^2, \quad \forall x \in \mathbb{R}^d, \qquad \text{for some } \mu > 0. \tag{1.4.3}$$

Definition 1.4.1 (Convexity)**.** *A function $f : \mathbb{R}^d \to \mathbb{R}$ is said to be convex if, $\forall x, y \in \mathbb{R}^d$ and $\lambda \in [0, 1]$, it satisfies*

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda) f(y). \tag{1.4.4}$$

If $f : \mathbb{R}^d \to \mathbb{R}$ is convex and differentiable, then

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle, \quad \forall x, y \in \mathbb{R}^d. \tag{1.4.5}$$

Definition 1.4.2 (Strong Convexity)**.** *A function $f : \mathbb{R}^d \to \mathbb{R}$ is said to be μ-strongly convex for some $\mu > 0$, if for any $x, y \in \mathbb{R}^d$ and $\lambda \in [0, 1]$:*

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda) f(y) - \frac{\mu}{2} \lambda (1 - \lambda) \|x - y\|^2. \tag{1.4.6}$$

Here, the constant μ is called the strong convexity parameter of f. Equivalently, $f : \mathbb{R}^d \to \mathbb{R}$ is μ-strongly convex if and only if there exists a convex function $g : \mathbb{R}^d \to \mathbb{R}$ such that

$$f(x) = g(x) + \frac{\mu}{2} \|x\|^2.$$

Observe that if $f : \mathbb{R}^d \to \mathbb{R}$ is μ-strongly convex and differentiable, then

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{\mu}{2} \|y - x\|^2, \quad \forall x, y \in \mathbb{R}^d. \tag{1.4.7}$$

Any strongly convex function admits a unique minimiser.

Definition 1.4.3 (Lipschitz continuity)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable. We say f is L-Lipschitz continuous if*

$$|f(x) - f(y)| \leq L \|x - y\|, \quad \forall x, y \in \mathbb{R}^d. \tag{1.4.8}$$

We say that a differentiable function f is *L-smooth* if its gradient ∇f is L-Lipschitz.

Strong Convexity and the Stability of the Mean Field In optimisation-based stochastic approximation, the mean field is often given by $h(x) = \nabla F(x)$ for some objective function F. If F is μ-strongly convex, then its gradient satisfies

$$\langle \nabla F(x), x - x^* \rangle \geq \mu \|x - x^*\|^2, \qquad \text{where } x^* \text{ is the unique minimiser of } F.$$

Consequently, the stability condition (1.4.3) holds, linking strong convexity of the objective to stability of the stochastic approximation recursion.

Let $R : \mathbb{R}^d \to (-\infty, +\infty]$ be a proper, lower semicontinuous, convex function. The proximal operator associated with R is defined by

$$\operatorname{prox}_{\eta R}(x) = \arg\min_{z \in \mathbb{R}^d} \left\{ R(z) + \frac{1}{2\eta} \|z - x\|^2 \right\} \qquad \text{for} \quad \eta > 0. \tag{1.4.9}$$

Since the quadratic term is $\frac{1}{\eta}$-strongly convex, the minimiser is unique and $\operatorname{prox}_{\eta R}(x)$ is well-defined. The proximal operator is characterised by the optimality condition

$$\frac{1}{\eta}(x - \operatorname{prox}_{\eta R}(x)) \in \partial R(\operatorname{prox}_{\eta R}(x)),$$

where ∂R denotes the subdifferential of R.

1.5 Standing Assumptions

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_k)_{k \geq 0}, \mathbb{P})$, and all conditional expectations are taken with respect to the filtration $(\mathcal{F}_k)$ unless stated otherwise.

Notation 1.5.1. *Vectors are denoted by lowercase letters (e.g. $x \in \mathbb{R}^d$). The Euclidean inner product and norm are written as $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$, respectively. For a function $F : \mathbb{R}^d \to \mathbb{R}$, we denote by x^* a minimiser of F when it exists.*

Stochastic Approximation Assumptions. Unless explicitly stated otherwise, stochastic approximation schemes are assumed to satisfy the recursion

$$x_{k+1} = x_k - \alpha_k \big(h(x_k) + \xi_{k+1} \big),$$

under the following conditions:

- The step-size sequence $(\alpha_k)_{k \geq 0}$ satisfies $\sum_k \alpha_k = \infty$ and $\sum_k \alpha_k^2 < \infty$.
- The noise sequence $(\xi_{k+1})_{k \geq 0}$ forms a martingale difference sequence with respect to $(\mathcal{F}_k)$, i.e. $\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = 0$ almost surely.
- There exists $\sigma^2 > 0$ such that $\mathbb{E}[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k] \leq \sigma^2$ almost surely.
- The drift $h : \mathbb{R}^d \to \mathbb{R}^d$ is Lipschitz continuous and satisfies the stability condition (1.4.3).

Optimisation Assumptions. When considering optimisation problems, we focus on objectives of the form

$$P(x) = F(x) + R(x),$$

where $F : \mathbb{R}^d \to \mathbb{R}$ is differentiable and $R : \mathbb{R}^d \to (-\infty, +\infty]$ is proper, lower semicontinuous, and convex. Unless stated otherwise, F is assumed to be μ-strongly convex and L-smooth.

Lemma 1.5.2 (Robbins–Siegmund [8])**.** *Let $(\Omega, \mathcal{F}, (\mathcal{F}_k)_{k\geq 0}, \mathbb{P})$ be a filtered probability space. Let V_k, γ_k, and ζ_k be non-negative $\mathcal{F}_k$-measurable random variables satisfying:*

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq (1 + \gamma_k)V_k + \zeta_k \quad a.s. \tag{1.5.1}$$

If $\sum_{k=0}^{\infty} \gamma_k < \infty$ and $\sum_{k=0}^{\infty} \zeta_k < \infty$ almost surely, then V_k converges almost surely to a finite random variable V_∞.

A corollary Lemma 1.5.2 for optimisation is the case of "contraction" where:

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq (1 - \alpha_k)V_k + \beta_k.$$

If $\sum \alpha_k = \infty$ (infinite contraction) and $\sum \beta_k < \infty$ (finite noise), then $V_\infty = 0$ almost surely.

Chapter 2

Proximal Gradient Methods

The aim of this chapter is to bridge classical stochastic approximation theory with modern stochastic optimisation algorithms, focusing on stochastic gradient and proximal methods. We have three specific goals in this chapter. **First** is to show that Robbins–Monro is a special case of the SA recursion in (1.4.1) with different choices of mean fields (h), noise (ξ_k), and measurable information ($\mathcal{F}_k$). **Second** is to show that stochastic gradient descent is stochastic approximation for finite-sum optimisation. **Third** is to show that proximal updates arise naturally for composite objectives $P(x) = F(x) + R(x)$.

2.1 Robbins–Monro as Stochastic Approximation

In the classical Robbins–Monro problem, we aim to find a point $x^* \in \mathbb{R}^d$ such that $h(x^*) = 0$, where $h : \mathbb{R}^d \to \mathbb{R}^d$ is an unknown function. The function h cannot be evaluated directly. Instead, at each iteration $k \geq 0$, one observes a noisy measurement of the form

$$Y_{k+1}(x) = h(x) + \varepsilon_{k+1}(x)$$

where the noise satisfies the martingale difference condition $\mathbb{E}[\varepsilon_{k+1}(x_k) \mid \mathcal{F}_k] = 0$. In the algorithm, the noise is evaluated at the current iterate, so the SA noise term is defined by

$$\xi_{k+1} := \varepsilon_{k+1}(x_k)$$

The Robbins–Monro update rule is given by

$$x_{k+1} = x_k - \alpha_k Y_{k+1}(x_k) \tag{2.1.1}$$

where $(\alpha_k)_{k\geq 0}$ is a sequence of positive step sizes. Substituting the observation model into (2.1.1):

$$x_{k+1} = x_k - \alpha_k\big(h(x_k) + \varepsilon_{k+1}(x_k)\big). \tag{2.1.2}$$

This coincides with the stochastic approximation recursion (1.4.1) where the noise term is $\xi_{k+1} = \varepsilon_{k+1}(x_k)$.

Theorem 2.1.1 (Robbins–Monro Convergence)**.** *Let $(x_k)_{k\geq 0}$ be the sequence of iterates generated by the Robbins–Monro algorithm (Equation 2.1.2). Under the stochastic approximation assumptions defined in Section 1.5, the sequence converges to the true root x^* almost surely:*

$$x_k \to x^* \quad a.s. \text{ as } k \to \infty.$$

Proof. We define the error $e_k := x_k - x^*$ and the squared error $V_k := \|e_k\|^2$. Subtracting x^* from both sides of the update rule yields:

$$e_{k+1} = e_k - \alpha_k\big(h(x_k) + \varepsilon_{k+1}(x_k)\big).$$

Taking the squared norm of both sides, we expand V_{k+1}:

$$\begin{aligned} V_{k+1} &= \|e_k - \alpha_k\big(h(x_k) + \varepsilon_{k+1}(x_k)\big)\|^2 \\ &= V_k - 2\alpha_k\langle e_k, h(x_k)\rangle - 2\alpha_k\langle e_k, \varepsilon_{k+1}(x_k)\rangle + \alpha_k^2\|h(x_k) + \varepsilon_{k+1}(x_k)\|^2. \end{aligned}$$

Now, we take the conditional expectation with respect to the filtration $\mathcal{F}_k$. Since x_k (and thus e_k and $h(x_k)$) are $\mathcal{F}_k$-measurable, we can pull them out of the expectation. By the martingale difference assumption, $\mathbb{E}[\varepsilon_{k+1}(x_k) \mid \mathcal{F}_k] = 0$. This eliminates the cross-term involving the noise:

$$\mathbb{E}[\langle e_k, \varepsilon_{k+1}(x_k)\rangle \mid \mathcal{F}_k] = \langle e_k, \mathbb{E}[\varepsilon_{k+1}(x_k) \mid \mathcal{F}_k]\rangle = 0.$$

Similarly, expanding the squared norm of the step gives a cross-term of $2\langle h(x_k), \varepsilon_{k+1}(x_k)\rangle$, which also vanishes under expectation. Thus:

$$\mathbb{E}[\|h(x_k) + \varepsilon_{k+1}(x_k)\|^2 \mid \mathcal{F}_k] = \|h(x_k)\|^2 + \mathbb{E}[\|\varepsilon_{k+1}(x_k)\|^2 \mid \mathcal{F}_k].$$

Using the bounded variance assumption $\mathbb{E}[\|\varepsilon_{k+1}(x_k)\|^2 \mid \mathcal{F}_k] \leq \sigma^2$, we obtain the bound:

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq V_k - 2\alpha_k\langle e_k, h(x_k)\rangle + \alpha_k^2\big(\|h(x_k)\|^2 + \sigma^2\big).$$

Next, we apply the stability condition $\langle x - x^*, h(x)\rangle \geq \mu\|x - x^*\|^2$, which implies

$$-2\alpha_k\langle e_k, h(x_k)\rangle \leq -2\mu\alpha_k V_k.$$

Substituting this into our inequality yields:

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq (1 - 2\mu\alpha_k)V_k + \alpha_k^2\big(\|h(x_k)\|^2 + \sigma^2\big).$$

Assuming h is L-Lipschitz and $h(x^*) = 0$, we have $\|h(x_k)\| \leq L\|e_k\|$, and thus $\|h(x_k)\|^2 \leq L^2 V_k$. This allows us to bound the expectation entirely in terms of V_k:

$$\begin{aligned} \mathbb{E}[V_{k+1} \mid \mathcal{F}_k] &\leq (1 - 2\mu\alpha_k)V_k + \alpha_k^2 L^2 V_k + \alpha_k^2\sigma^2 \\ &= (1 - 2\mu\alpha_k + \alpha_k^2 L^2)V_k + \alpha_k^2\sigma^2. \end{aligned}$$

Recall that the step sizes satisfy $\alpha_k \to 0$. We want the coefficient of V_k to reflect a strict contraction. Specifically, we require $1 - 2\mu\alpha_k + \alpha_k^2 L^2 \leq 1 - \mu\alpha_k$, which simplifies to $\alpha_k \leq \frac{\mu}{L^2}$. Since $\alpha_k \to 0$, there exists an index k_0 such that for all $k \geq k_0$, this condition holds. For such sufficiently large k, the inequality simplifies to:

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq (1 - \mu\alpha_k)V_k + \alpha_k^2\sigma^2.$$

To apply Theorem 1.5.2, we define the non-negative sequences $a_k := \mu\alpha_k$ and $b_k := \sigma^2\alpha_k^2$. This puts our inequality into the canonical form:

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq (1 - a_k)V_k + b_k.$$

We verify the conditions for Theorem 1.5.2. First, since the step sizes satisfy $\sum \alpha_k^2 < \infty$, the noise term is summable:

$$\sum_{k=0}^{\infty} b_k = \sigma^2 \sum_{k=0}^{\infty} \alpha_k^2 < \infty.$$

Second, since the step sizes satisfy $\sum \alpha_k = \infty$, the contraction coefficients sum to infinity:

$$\sum_{k=0}^{\infty} a_k = \mu \sum_{k=0}^{\infty} \alpha_k = \infty.$$

By Theorem 1.5.2, the sequence V_k converges almost surely to a finite random variable V_∞, and the weighted sum of the sequence is finite almost surely:

$$\sum_{k=0}^{\infty} a_k V_k < \infty \quad \text{a.s.}$$

We conclude that $V_\infty = 0$ almost surely. Suppose, for the sake of contradiction, that $V_k \to V_\infty > 0$ with positive probability. Then, for sufficiently large k, we would have $V_k \geq \frac{V_\infty}{2}$. This would imply:

$$\sum_{k=0}^{\infty} a_k V_k \geq \frac{V_\infty}{2} \sum_{k=0}^{\infty} a_k = \infty,$$

which contradicts the finite sum property. Therefore, $V_\infty = 0$. Since $V_k = \|x_k - x^*\|^2$, it immediately follows that $x_k \to x^*$ almost surely. □

Robbins–Monro can be used as a general template for understanding stochastic gradient–type algorithms. It shows that convergence follows from a stochastic approximation structure characterised by a stable mean field (which provides the drift towards the optimum) and unbiased noise. The analysis highlights the fundamental tension in stochastic optimisation: the drift term $(-\mu\alpha_k V_k)$ drives the iterate towards x^*, while the noise term $(\sigma^2\alpha_k^2)$ pushes it away. Convergence is only achieved because the noise variance decays faster $(\sum \alpha_k^2 < \infty)$ than the drift accumulates $(\sum \alpha_k = \infty)$.

Because the noise variance σ^2 remains non-zero at the optimum, satisfying the summable-noise condition requires a diminishing step-size schedule $(\alpha_k \to 0)$. This necessity is what fundamentally limits classical SGD to a sublinear convergence rate. Consequently, if a later

algorithm in this report can be written in Robbins–Monro form, its convergence may be analysed using these same arguments. However, if an algorithm can eliminate the variance at the optimum (as in variance-reduced methods), it may bypass the requirement for diminishing step sizes, potentially unlocking faster linear convergence rates.

2.2 SGD as Stochastic Approximation for Empirical Risk Minimisation

We now aim to show that stochastic gradient descent (SGD) is an instance of the stochastic approximation recursion (1.4.1). Specifically, we express SGD as

$$x_{k+1} = x_k - \alpha_k\big(h(x_k) + \xi_{k+1}\big),$$

by identifying the mean field as the full gradient $h(x) = \nabla F(x)$ and the noise term as the sampling error $\xi_{k+1} = \nabla f_{i_{k+1}}(x_k) - \nabla F(x_k)$. This makes the Robbins–Monro convergence mechanism from Section 1.5 directly applicable to SGD: the drift term $\nabla F(x_k)$ drives the iterates towards the minimiser, while the stochastic gradient noise determines what step-size conditions are needed for convergence.

Remark. Throughout this section, we assume (1) a μ-strongly convex, L-smooth objective F, and (2) that the gradient ∇F satisfies the stability condition (1.4.3).

ERM Setup Let $\{z_1, \dots, z_n\}$ denote a finite dataset and define the per-sample loss functions

$$f_i(x) := \ell(x; z_i), \qquad i = 1, \dots, n.$$

The empirical risk minimisation (ERM) problem is then given by

$$\min_{x \in \mathbb{R}^d} F(x), \qquad F(x) := \frac{1}{n}\sum_{i=1}^{n} f_i(x).$$

At iteration k, an index $i_{k+1} \in \{1, \dots, n\}$ is sampled uniformly at random, independently of the past iterates. The stochastic gradient is defined as $\nabla f_{i_{k+1}}(x_k)$, and the SGD update rule takes the form

$$x_{k+1} = x_k - \alpha_k \nabla f_{i_{k+1}}(x_k),$$

where $(\alpha_k)_{k \geq 0}$ is a sequence of positive step sizes.

Identification with the SA Recursion We now express the SGD update in stochastic approximation form. We define the mean field and the noise sequence as mentioned above:

$$h(x) := \nabla F(x) \quad \text{and} \quad \xi_{k+1} := \nabla f_{i_{k+1}}(x_k) - \nabla F(x_k).$$

Substituting these definitions into the stochastic approximation recursion yields

$$\begin{aligned} x_{k+1} &= x_k - \alpha_k\big(h(x_k) + \xi_{k+1}\big) \\ &= x_k - \alpha_k\big(\nabla F(x_k) + \nabla f_{i_{k+1}}(x_k) - \nabla F(x_k)\big) = x_k - \alpha_k \nabla f_{i_{k+1}}(x_k), \end{aligned}$$

which coincides exactly with the SGD update rule. Hence, stochastic gradient descent is a special case of the stochastic approximation recursion (1.4.1), with mean field given by the full gradient ∇F and noise induced by random sampling.

Unbiasedness of the Stochastic Gradient A key requirement of the stochastic approximation framework is that the noise sequence $(\xi_{k+1})_{k\geq 0}$ satisfies $\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = 0$ almost surely. In the ERM setting, i_{k+1} is sampled uniformly from $\{1, \ldots, n\}$ independently of $\mathcal{F}_k$, and x_k is $\mathcal{F}_k$-measurable. Conditioning on $\mathcal{F}_k$ therefore fixes x_k:

$$\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = \mathbb{E}[\nabla f_{i_{k+1}}(x_k) \mid \mathcal{F}_k] - \nabla F(x_k).$$

By uniform sampling,

$$\mathbb{E}[\nabla f_{i_{k+1}}(x_k) \mid \mathcal{F}_k] = \frac{1}{n}\sum_{i=1}^{n} \nabla f_i(x_k) = \nabla F(x_k),$$

so $\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = 0$ almost surely.

Variance of the Stochastic Gradient The stochastic gradient is unbiased but has non-zero variance, a fundamental obstacle to using a constant step size in classical SGD. This is why Robbins–Monro requires $\sum_k \alpha_k^2 < \infty$. To formalise this, we first define the conditional second moment of the noise term:

$$\mathbb{E}\big[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k\big].$$

Since x_k is $\mathcal{F}_k$-measurable, conditioning on $\mathcal{F}_k$ fixes x_k, and the only remaining randomness is the fresh sample i_{k+1}. Classical stochastic approximation typically bounds this quantity uniformly:

Assumption 1 (Classical Bounded Conditional Variance [2])**.** *There exists a constant $\sigma^2 > 0$ such that for all $k \geq 0$,*

$$\mathbb{E}\left[\|\nabla f_{i_{k+1}}(x_k) - \nabla F(x_k)\|^2 \mid \mathcal{F}_k\right] \leq \sigma^2 \quad \textit{almost surely.} \tag{2.2.1}$$

This classical assumption is widely used in the introductory literature to establish convergence rates for stochastic gradient methods (e.g., Assumption 2.3.2 in [2]).

Remark (The Quadratic Contradiction and Bounded Domains)*.* While Assumption 1 is a standard theoretical starting point, it strictly fails for strongly convex quadratic objectives (which we analyse in Chapter 4) on an unbounded domain. Because the gradients of quadratics grow linearly with $\|x\|$, the variance cannot be globally bounded.

Therefore, to align with our experimental setup, we adopt the modern standard practice (e.g., Bottou et al., 2018 [4]) of replacing the global bound with a trajectory bound. We assume the algorithm generates iterates that remain almost surely within a compact level set. By the continuity of the gradients, this ensures the conditional variance remains uniformly bounded by some $\sigma^2 < \infty$ strictly along the algorithmic trajectory.

Crucially, even if we bound the trajectory, a defining characteristic of classical SGD is that the noise generally does not vanish at the exact minimiser x^* of F. Recall that the noise at the optimum is given by:

$$\xi_{k+1}(x^*) = \nabla f_{i_{k+1}}(x^*) - \nabla F(x^*).$$

In the strongly convex case, the full gradient at the optimum is $\nabla F(x^*) = 0$, which leaves:

$$\xi_{k+1}(x^*) = \nabla f_{i_{k+1}}(x^*).$$

Unless every individual component gradient vanishes at the exact same point ($\nabla f_i(x^*) = 0 \quad \forall i$), the variance at the optimum is strictly positive:

$$\sigma_*^2 := \mathbb{E}[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k]\Big|_{x_k=x^*} = \frac{1}{n}\sum_{i=1}^{n} \|\nabla f_i(x^*)\|^2 > 0. \tag{2.2.2}$$

Hence, in general, SGD has a persistent noise floor σ_*^2 at the optimum. The overall variance may be bounded, but it does not decay to zero as $x_k \to x^*$.

Because the stochastic gradient noise maintains this non-zero variance at x^*, taking a constant step size $\alpha_k \equiv \alpha$ leads to a steady-state error of order α rather than exact convergence. This explains exactly why the Robbins–Monro condition $\sum_k \alpha_k^2 < \infty$ (and hence $\alpha_k \to 0$) is imposed in classical SA analysis: it forces the cumulative contribution of the noise to be finite, allowing convergence to the exact solution.

Interpretation The stochastic approximation interpretation of SGD clarifies both why the algorithm converges under diminishing step sizes and why its convergence rate is limited.

From Section 2.2, the stochastic gradient can be decomposed as

$$\nabla f_{i_{k+1}}(x_k) = \nabla F(x_k) + \xi_{k+1},$$

where the noise sequence $(\xi_{k+1})_{k\geq 0}$ is unbiased but has non-vanishing variance at the x^*. Consequently, SGD exactly matches the Robbins–Monro SA recursion with mean field $h(x) = \nabla F(x)$ and persistent noise.

The Robbins–Monro convergence proof in Section 2.1 shows that convergence is driven by a balance between two competing effects:

- the drift term, induced by strong convexity of F, which brings the error toward the minimiser, and
- the noise term, arising from stochastic gradient sampling, which adds variance at every iteration.

Because the noise variance does not vanish as $x_k \to x^*$, exact convergence is only possible if the step sizes decay sufficiently fast so that the cumulative noise contribution $\sum_k \alpha_k^2$ is finite. This is precisely the Robbins–Monro condition $\sum_k \alpha_k^2 < \infty$, and it explains why classical SGD requires diminishing step sizes.

At the same time, the condition $\sum_k \alpha_k = \infty$ ensures that the algorithm keeps making cumulative progress, rather than taking steps that become too small too quickly. These two conditions explain the behaviour of SGD: almost sure convergence with sublinear rates under diminishing step sizes.

This perspective also reveals the fundamental limitation of classical SGD. As long as the noise variance remains non-zero at the optimum, constant step sizes cannot yield exact convergence. Overcoming this limitation requires modifying the noise structure, a principle that motivates variance-reduced methods.

2.3 From Gradient Descent to Proximal Gradient Methods

We now show that proximal gradient methods arise naturally when stochastic approximation is applied to composite objectives of the form

$$P(x) = F(x) + R(x), \tag{2.3.1}$$

where $F : \mathbb{R}^d \to \mathbb{R}$ is smooth and $R : \mathbb{R}^d \to (-\infty, +\infty]$ is proper, convex, and possibly non-smooth.

Why Proximal Methods are Needed The stochastic approximation and stochastic gradient methods discussed so far assume a smooth objective. However, many optimisation problems of interest are composite, as in (2.3.1), and the non-smooth term R is often essential for modelling regularisation or constraints.

Typical examples include:

- ℓ_1-regularisation, $R(x) = \lambda\|x\|_1$, used to promote sparsity,
- indicator functions $R(x) = \mathbb{I}_{\mathcal{C}}(x)$ encoding hard constraints.

In such cases, $\nabla R(x)$ may not exist, so a direct gradient descent step on P is not well-defined. Even so, the optimisation problem remains convex and well posed, and one would like to retain the scalability and simplicity of first-order methods.

This motivates proximal methods, which replace an explicit gradient step on R with an implicit update defined through the proximal operator. Proximal algorithms therefore extend gradient-based methods to composite objectives while preserving a first-order structure.

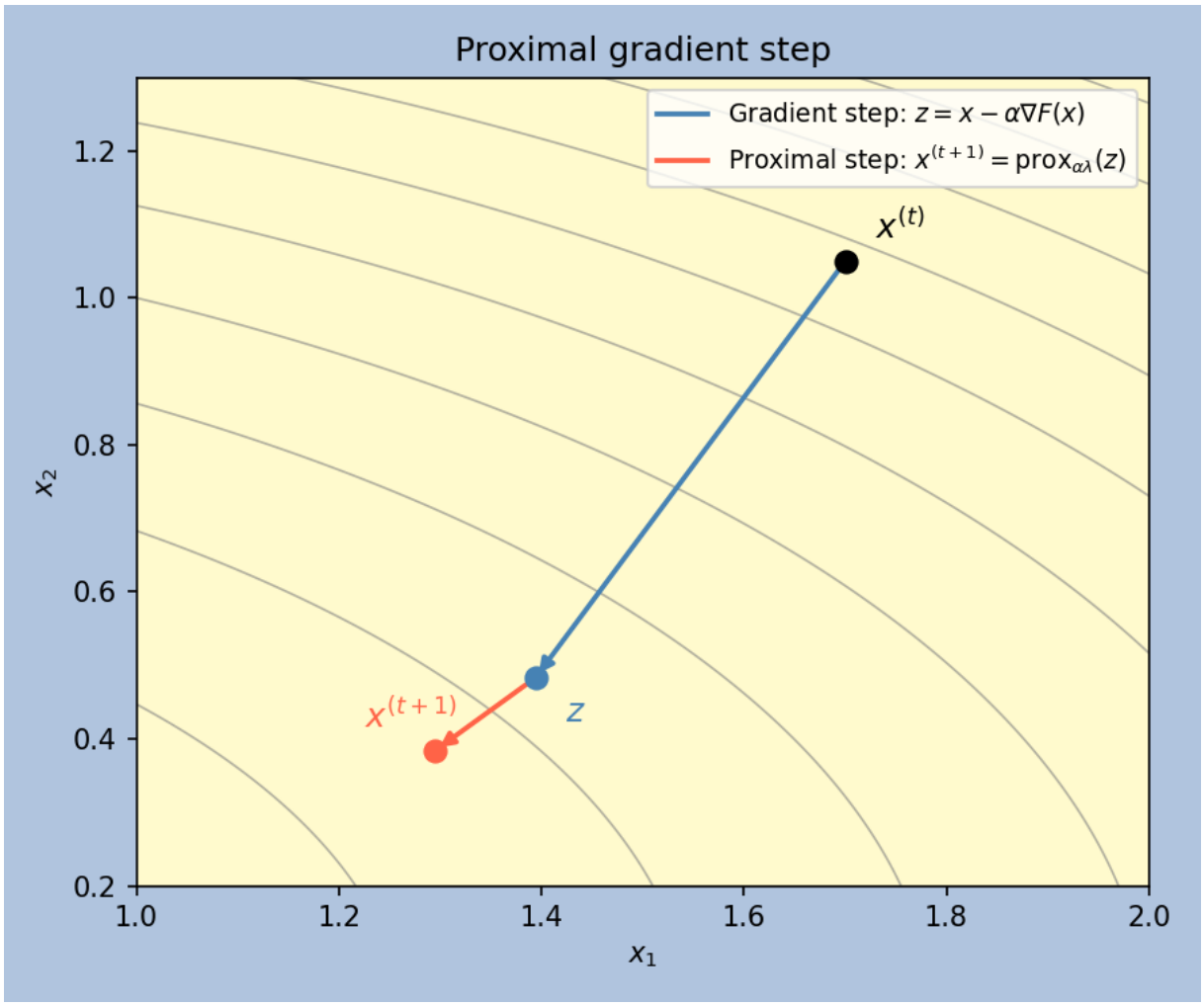

Figure 2.1: Shows an illustration of a proximal gradient step. A standard gradient step (blue) moves the iterate in the direction of $-\nabla F(x)$. The proximal ℓ_1 operator then "snaps" the point toward coordinate axes, promoting sparsity. This geometric interpretation clarifies the forward–backward structure of proximal methods.

Deterministic Proximal Gradient Method Given (2.3.1), a normal gradient descent step on P would require evaluating $\nabla R(x)$, which, as stated previously, may not exist. Proximal gradient methods instead apply a gradient step to F followed by a proximal correction for R. For a step size $\alpha > 0$, the update is

$$x_{k+1} = \mathrm{prox}_{\alpha R}(x_k - \alpha\nabla F(x_k)) \,.$$

This can be interpreted in the following way: first, a standard gradient step is taken with respect to the smooth part F. Then, the proximal operator accounts for the non-smooth structure of R while staying within a first-order framework.

Proximal Gradient as SA We now show that deterministic proximal gradient descent can be written in a stochastic-approximation-like recursion. Although no randomness is present here, this viewpoint is useful because it makes the transition to Prox-SGD and variance-reduced proximal methods easier.

Recall the proximal gradient update

$$x_{k+1} = \mathrm{prox}_{\alpha R}(x_k - \alpha \nabla F(x_k)) \,.$$

Using the definition of the proximal operator, this is equivalently

$$x_{k+1} = \arg\min_{z \in \mathbb{R}^d} \left\{ \langle \nabla F(x_k), z - x_k \rangle + R(z) + \frac{1}{2\alpha} \|z - x_k\|^2 \right\}. \tag{2.3.2}$$

The first-order optimality condition for (2.3.2) is

$$0 \in \nabla F(x_k) + \frac{1}{\alpha}(x_{k+1} - x_k) + \partial R(x_{k+1}). \tag{2.3.3}$$

Equivalently, there exists a $v_{k+1} \in \partial R(x_{k+1})$ such that

$$x_{k+1} = x_k - \alpha\big(\nabla F(x_k) + v_{k+1}\big), \qquad v_{k+1} \in \partial R(x_{k+1}). \tag{2.3.4}$$

Equation (2.3.4) has the same recursion structure as stochastic approximation: an iterate update driven by a drift term. The key difference is that the non-smooth component is not handled via $\nabla R(x_k)$, but via an implicit selection $v_{k+1} \in \partial R(x_{k+1})$ produced by the proximal step. In particular, the drift object here is the monotone operator $\nabla F + \partial R$, whose root corresponds to the first-order condition $0 \in \nabla F(x^*) + \partial R(x^*)$.

Unlike the stochastic approximation recursion studied earlier, (2.3.4) contains no stochastic noise term. However, it shares the same drift-based structure: the iterate moves according to a stable first-order condition, with stability inherited from strong convexity of the composite objective $P(x) = F(x) + R(x)$.

Proximal Stochastic Gradient Descent We now move to proximal stochastic gradient descent (Prox-SGD), which combines stochastic gradient updates with proximal operators to handle composite objectives of the form (2.3.1). Assume the finite-sum structure

$$F(x) = \frac{1}{n} \sum_{i=1}^{n} f_i(x),$$

and that at iteration k an index $i_{k+1} \in \{1, \ldots, n\}$ is sampled uniformly at random. The Prox-SGD update is

$$x_{k+1} = \mathrm{prox}_{\alpha_k R}\big(x_k - \alpha_k \nabla f_{i_{k+1}}(x_k)\big) \,, \tag{2.3.5}$$

where $(\alpha_k)_{k \geq 0}$ is a sequence of positive step sizes.

This is the stochastic equivalent of deterministic proximal gradient descent: a stochastic gradient step is taken on the smooth component F, followed by a proximal correction for the non-smooth component R.

To express (2.3.5) in an SA-like recursion, we reuse the standard SGD sampling noise

$$\xi_{k+1} := \nabla f_{i_{k+1}}(x_k) - \nabla F(x_k).$$

Using the proximal optimality condition, there exists $v_{k+1} \in \partial R(x_{k+1})$ such that the Prox-SGD iterate satisfies

$$x_{k+1} = x_k - \alpha_k\big(\nabla F(x_k) + \xi_{k+1} + v_{k+1}\big), \qquad v_{k+1} \in \partial R(x_{k+1}). \tag{2.3.6}$$

Equation (2.3.6) shows that Prox-SGD follows the same stochastic approximation structure as Robbins–Monro: the update consists of a drift term (now governed by the composite first-order condition) plus a martingale-difference noise term induced by sampling.

Remark. Strictly speaking, (2.3.6) represents an *implicit* stochastic approximation because v_{k+1} is evaluated at the new iterate x_{k+1} and is therefore not $\mathcal{F}_k$-measurable. However, because the proximal operator is firmly non-expansive (1-Lipschitz), the update remains stable. A fully rigorous convergence proof requires leveraging this non-expansiveness rather than relying solely on the explicit Robbins–Monro framework.

Noise Behaviour in Proximal Methods Although Prox-SGD fits naturally into the stochastic approximation framework via (2.3.6), its convergence behaviour is still limited by the noise from stochastic sampling. The proximal operator changes how the non-smooth term R is handled, but it does not remove the stochasticity coming from estimating $\nabla F(x_k)$ with $\nabla f_{i_{k+1}}(x_k)$.

Let x^* denote the unique minimiser of $P(x) = F(x) + R(x)$. At the optimum, the composite first-order condition is

$$0 \in \nabla F(x^*) + \partial R(x^*).$$

However, the sampling noise in (2.3.6) becomes

$$\xi_{k+1} = \nabla f_{i_{k+1}}(x^*) - \nabla F(x^*),$$

which is generally non-zero unless all component gradients coincide at x^*. In particular, in the finite-sum setting the conditional variance of the sampling noise does not vanish at the solution:

$$\mathbb{E}[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k]\Big|_{x_k = x^*} > 0.$$

This implies that the noise level remains bounded away from zero, i.e. there exists $c > 0$ such that

$$\mathbb{E}[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k] \geq c \quad \text{for } x_k \text{ sufficiently close to } x^*.$$

As shown in the Robbins–Monro convergence analysis, persistent noise of this form forces diminishing step sizes. We require

$$\sum_{k=0}^{\infty} \alpha_k^2 < \infty, \qquad \alpha_k \to 0,$$

so that the cumulative contribution of the noise remains finite. Consequently, Prox-SGD inherits the same limitation as classical SGD: diminishing step sizes lead to sublinear convergence rates, even under strong convexity.

This limitation is not caused by the proximal operator. It is caused by the fact that the stochastic gradient estimator for ∇F has non-vanishing variance at the solution. Overcoming this requires modifying the noise structure itself, which is the purpose of variance-reduced methods.

Connection to Robbins–Monro We now make the connection between proximal stochastic gradient methods and the classical Robbins–Monro stochastic approximation framework. Recall the canonical recursion

$$x_{k+1} = x_k - \alpha_k\big(h(x_k) + \xi_{k+1}\big), \tag{2.3.7}$$

where the mean field is stable, the noise sequence is a martingale difference, and the step sizes satisfy $\sum \alpha_k = \infty$ and $\sum \alpha_k^2 < \infty$.

Comparing (2.3.6) with (2.3.7), Prox-SGD has the same stochastic approximation structure, but with a drift determined by the composite first-order condition. Specifically, the root is characterised by

$$0 \in \nabla F(x^*) + \partial R(x^*),$$

and the drift is represented through the implicit selection $v_{k+1} \in \partial R(x_{k+1})$.

Under the standing assumptions that F is μ-strongly convex and R is convex, the composite objective $P(x) = F(x) + R(x)$ is strongly convex and admits a unique minimiser x^*. Furthermore, strong convexity implies a drift toward x^*, which plays the same stabilising role as the stability condition in the Robbins–Monro analysis.

The Robbins–Monro mechanism therefore applies directly to (2.3.6). Convergence is driven by the competition between:

- the stabilising drift toward x^*, caused by strong convexity of the composite objective, and
- the stochastic sampling noise, whose contribution scales like α_k^2 per iteration.

As shown in Section 2.1, almost sure convergence follows only when the accumulated noise remains finite, which enforces diminishing step sizes $\alpha_k \to 0$. This requirement persists in the proximal setting because the sampling noise does not vanish at the optimum.

Consequently, proximal stochastic gradient methods remain within the classical Robbins–Monro umbrella. They inherit its convergence guarantees, but also its limitations. In particular, persistent noise prevents the use of constant step sizes and rules out linear convergence rates, even under strong convexity.

This motivates variance-reduced proximal methods, which modify the gradient estimator so that the stochastic noise vanishes at the solution, breaking free from the classical Robbins–Monro constraints.

2.4 Comparative Summary and Transition to Variance Reduction

This chapter has shown that a wide range of algorithms can be understood within a unified stochastic approximation (SA) framework. The key differences between methods arise not from

the update structure itself, but from the choice of mean field and, crucially, the behaviour of the noise.

- **Robbins–Monro** solves root-finding problems using unbiased but noisy observations. Convergence relies on a stable mean field and diminishing step sizes to control persistent noise.
- **Stochastic Gradient Descent (SGD)** is a direct instance of Robbins–Monro applied to empirical risk minimisation. The mean field is the full gradient, while noise arises from random data sampling and remains non-zero at the optimum.
- **Proximal Gradient Methods** extend first-order optimisation to composite objectives by handling non-smooth terms implicitly via the proximal operator. These methods preserve the SA structure while accommodating constraints and regularisation.
- **Proximal Stochastic Gradient Descent** combines stochastic gradients with proximal updates. Although stability is retained under strong convexity, the noise variance again does not vanish at the solution, enforcing diminishing step sizes.

Across all classical SA methods studied in this chapter, convergence is achieved through the same mechanism: accumulated contraction from a stable mean field, balanced against the cumulative effect of stochastic noise. When the noise remains active at the solution, exact convergence is only achievable with diminishing step sizes, since the Robbins–Monro condition $\sum_k \alpha_k^2 < \infty$ is needed to keep the total noise contribution finite. This is the reason classical SGD-type methods are limited to sublinear rates, even in strongly convex settings.

In this chapter, we showed that SGD and Prox-SGD naturally fit into the Robbins–Monro SA framework. Under strong convexity, the drift pulls the iterates towards the minimiser but the sampling noise does not disappear there. In particular, the conditional variance of the sampling error remains bounded away from zero near the solution. This persistent noise is what forces diminishing step sizes through the Robbins–Monro condition $\sum_k \alpha_k^2 < \infty$, and it is the structural reason classical stochastic gradient methods show at best sublinear convergence rates, even in strongly convex finite-sum problems.

Variance-reduced methods are designed to modify this aspect of the recursion. They exploit the finite-sum structure $F(x) = \frac{1}{n}\sum_{i=1}^n f_i(x)$ to build unbiased control-variate gradient estimators whose variance diminishes as the iterates approach the minimiser. In stochastic approximation terms, the mean field is unchanged (the full gradient in the smooth case, or the composite first-order condition in the proximal case), but the noise process becomes iterate-dependent and asymptotically vanishing. The aim of this chapter is to make this change precise for SVRG and Prox-SVRG and to explain why it permits constant step sizes and enables linear convergence under strong convexity. This brings us to Chapter 3.

Chapter 3

Variance-Reduced Methods

We now explore what changes in the stochastic approximation recursion when moving from classical SGD to variance-reduced methods, and how this altered noise behaviour drives faster convergence. Recall that variance-reduced methods rely on the finite-sum structure

$$F(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x),$$

where each f_i corresponds to an individual sample or data point. This structure provides information that is not available in generic stochastic approximation. **First**, the full gradient is well-defined and computable:

$$\nabla F(x) = \frac{1}{n}\sum_{i=1}^{n} \nabla f_i(x),$$

and **second**, one can form differences $\nabla f_i(x) - \nabla f_i(y)$ that cancel a large part of the sampling variability, because the same component functions reappear across iterations. This is the basis of control variates.

Variance-reduced algorithms exploit these two facts by periodically computing $\nabla F(\tilde{x})$ at a reference point $\tilde{x}$ and correcting stochastic gradients using component-wise differences. SVRG and Prox-SVRG implement this idea in their gradient estimator, which is the starting point for the noise analysis in the remainder of the chapter.

3.1 Stochastic Variance-Reduced Gradient (SVRG)

Stochastic Variance-Reduced Gradients (SVRG) [6] operate in epochs. At the beginning of each epoch, a reference point $\tilde{x}$ is fixed and the full gradient at this point is computed:

$$\nabla F(\tilde{x}) = \frac{1}{n}\sum_{i=1}^{n} \nabla f_i(\tilde{x}).$$

Within the epoch, stochastic updates are performed using a variance-reduced gradient estimator. Specifically, at iteration k, an index $i_{k+1} \in \{1, \ldots, n\}$ is sampled uniformly at random, and the

gradient estimator is defined as

$$g_{k+1} = \nabla f_{i_{k+1}}(x_k) - \nabla f_{i_{k+1}}(\tilde{x}) + \nabla F(\tilde{x}). \tag{3.1.1}$$

The SVRG update rule then takes the form

$$x_{k+1} = x_k - \alpha\, g_{k+1}, \tag{3.1.2}$$

where $\alpha > 0$ is typically taken constant. The estimator (3.1.1) has two key properties. First, it is unbiased:

$$\mathbb{E}[g_{k+1} \mid \mathcal{F}_k] = \nabla F(x_k),$$

since $\mathbb{E}[\nabla f_{i_{k+1}}(x) \mid \mathcal{F}_k] = \nabla F(x)$ under uniform sampling. Second, its variance is controlled by the distance between x_k and the reference point $\tilde{x}$ through the difference term

$$\nabla f_{i_{k+1}}(x_k) - \nabla f_{i_{k+1}}(\tilde{x}).$$

When x_k remains close to $\tilde{x}$ within an epoch, this correction term is small and the overall variance of g_{k+1} is reduced. We summarise this procedure in Algorithm 1.

Algorithm 1 Stochastic Variance Reduced Gradient (SVRG)

Require: Initial point $\tilde{x}^{(0)} \in \mathbb{R}^d$, step size $\alpha > 0$, epoch length m

```
for epoch s = 0, 1, 2, ... do
    Set reference point x̃ = x̃^(s)
    Compute full gradient ∇F(x̃) = (1/n) Σ_{i=1}^n ∇f_i(x̃)
    Initialise x_0 = x̃
    for k = 0 to m − 1 do
        Sample index i_{k+1} ~ Unif{1, ..., n}
        g_{k+1} ← ∇f_{i_{k+1}}(x_k) − ∇f_{i_{k+1}}(x̃) + ∇F(x̃)
        x_{k+1} ← x_k − α g_{k+1}
    end for
    Set x̃^(s+1) = x_m
end for
```

3.2 Proximal Stochastic Variance Reduced Gradient (Prox-SVRG)

For composite optimisation problems of the form

$$P(x) = F(x) + R(x),$$

where F is smooth and R is convex but possibly non-smooth, SVRG can be combined with proximal updates to yield the Prox-SVRG algorithm [5]. In this setting, the variance-reduced gradient estimator g_{k+1} is defined as in (3.1.1), and the update takes the proximal form

$$x_{k+1} = \operatorname{prox}_{\alpha R}\big(x_k - \alpha\, g_{k+1}\big). \tag{3.2.1}$$

As in Prox-SGD, the proximal operator handles the non-smooth component R implicitly, while the smooth component F is handled by a gradient step. The difference lies in the structure of g_{k+1}. The control-variate correction reduces the sampling variability within each epoch, and this variability shrinks as both x_k and the reference point $\tilde{x}$ approach the minimiser. Algorithm 2 summarises the Prox-SVRG procedure.

Algorithm 2 Proximal Stochastic Variance Reduced Gradient (Prox-SVRG)

Require: Initial point $\tilde{x}^{(0)} \in \mathbb{R}^d$, step size $\alpha > 0$, epoch length m

```
for epoch s = 0, 1, 2, ... do
    Set reference point x̃ = x̃^(s)
    Compute full gradient ∇F(x̃) = (1/n) Σ_{i=1}^n ∇f_i(x̃)
    Initialise x_0 = x̃
    for k = 0 to m − 1 do
        Sample index i_{k+1} ~ Unif{1, ..., n}
        g_{k+1} ← ∇f_{i_{k+1}}(x_k) − ∇f_{i_{k+1}}(x̃) + ∇F(x̃)
        x_{k+1} ← prox_{αR}(x_k − α g_{k+1})
    end for
    Set x̃^(s+1) = x_m
end for
```

3.3 Classical vs Variance-Reduced SGD

Proposition 3.3.1 (Variance Reduction at the Optimum)**.** *Let x^* be the minimiser of the objective function F. In variance-reduced methods such as SVRG and Prox-SVRG, the stochastic gradient estimator g_{k+1} satisfies:*

$$\lim_{x_k, \tilde{x} \to x^*} \mathbb{E}\big[\|g_{k+1} - \nabla F(x_k)\|^2 \mid \mathcal{F}_k\big] = 0,$$

provided the reference point $\tilde{x}$ is updated such that it converges to x^*.

Proof. Conditioning on $\mathcal{F}_k$ fixes x_k and $\tilde{x}$, and the only randomness is the fresh index i_{k+1}. Write

$$\xi_{k+1} = \big(\nabla f_{i_{k+1}}(x_k) - \nabla f_{i_{k+1}}(\tilde{x})\big) - \Big(\nabla F(x_k) - \nabla F(\tilde{x})\Big).$$

Define: $\quad \Delta_i := \nabla f_i(x_k) - \nabla f_i(\tilde{x}) \quad , \quad \bar{\Delta} := \frac{1}{n}\sum_{i=1}^n \Delta_i = \nabla F(x_k) - \nabla F(\tilde{x})$.
Then $\xi_{k+1} = \Delta_{i_{k+1}} - \bar{\Delta}$, hence

$$\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = \mathbb{E}[\Delta_{i_{k+1}} \mid \mathcal{F}_k] - \bar{\Delta} = \bar{\Delta} - \bar{\Delta} = 0.$$

Using $\mathbb{E}\|Z - \mathbb{E}Z\|^2 \le \mathbb{E}\|Z\|^2$,

$$\mathbb{E}\big[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k\big] = \mathbb{E}\big[\|\Delta_{i_{k+1}} - \bar{\Delta}\|^2 \mid \mathcal{F}_k\big] \le \mathbb{E}\big[\|\Delta_{i_{k+1}}\|^2 \mid \mathcal{F}_k\big],$$

By L-Lipschitz continuity of each ∇f_i,

$$\|\Delta_{i_{k+1}}\| = \|\nabla f_{i_{k+1}}(x_k) - \nabla f_{i_{k+1}}(\tilde{x})\| \le L\|x_k - \tilde{x}\|,$$

so

$$\mathbb{E}\big[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k\big] \le L^2\|x_k - \tilde{x}\|^2.$$

Finally, by $\|x_k - \tilde{x}\| \le \|x_k - x^*\| + \|\tilde{x} - x^*\|$ and $(a+b)^2 \le 2a^2 + 2b^2$,

$$\|x_k - \tilde{x}\|^2 \le 2\|x_k - x^*\|^2 + 2\|\tilde{x} - x^*\|^2,$$

Therefore, as $x_k, \tilde{x} \to x^*$, the conditional variance tends to 0. □

Remark. In classical SGD, the stochastic gradient noise ξ_{k+1} maintains a strictly non-zero variance even at the minimiser x^*. The asymptotic vanishing of variance established in Theorem 3.3.1 is the primary mathematical mechanism that permits the use of constant step sizes, thereby enabling linear convergence rates under strong convexity.

SA Interpretation of Variance-Reduced Updates We now interpret variance-reduced methods within the stochastic approximation (SA) framework developed in Chapters 1 and 2. The point of this section is to isolate what changes from classical SGD: the drift (mean field) is the same, but the noise process is fundamentally different.

SVRG in SA form. Within an epoch, SVRG updates

$$x_{k+1} = x_k - \alpha\, g_{k+1},$$

with the variance-reduced estimator

$$g_{k+1} = \nabla f_{i_{k+1}}(x_k) - \nabla f_{i_{k+1}}(\tilde{x}) + \nabla F(\tilde{x}),$$

where $\tilde{x}$ is the fixed reference point for that epoch. Define the drift and noise by

$$h(x) := \nabla F(x), \qquad \xi_{k+1} := g_{k+1} - \nabla F(x_k).$$

Then SVRG takes the SA recursion

$$x_{k+1} = x_k - \alpha\big(h(x_k) + \xi_{k+1}\big). \tag{3.3.1}$$

Expanding the definition gives

$$\xi_{k+1} = \nabla f_{i_{k+1}}(x_k) - \nabla f_{i_{k+1}}(\tilde{x}) + \nabla F(\tilde{x}) - \nabla F(x_k).$$

Unbiasedness: Conditioning on $\mathcal{F}_k$ fixes x_k and $\tilde{x}$, and the only randomness is the fresh index i_{k+1}. By uniform sampling,

$$\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = 0,$$

so (3.3.1) is a Robbins–Monro-type recursion with the same mean field as SGD, but with a different noise term.

Lemma 3.3.2 (SVRG Variance Bound)**.** *Under the standing assumption that each component function ∇f_i is L-Lipschitz continuous, the conditional second moment of the stochastic gradient noise ξ_{k+1} satisfies:*

$$\mathbb{E}\big[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k\big] \le C\Big(\|x_k - x^*\|^2 + \|\tilde{x} - x^*\|^2\Big), \tag{3.3.2}$$

for some constant $C > 0$. Consequently, if $x_k \to x^*$ and the reference point $\tilde{x} \to x^*$, the noise variance strictly vanishes:

$$\lim_{x_k, \tilde{x} \to x^*} \mathbb{E}\big[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k\big] = 0.$$

Proof. The argument is the same variance-decomposition step used earlier for sampling noise. Condition on $\mathcal{F}_k$ so that only i_{k+1} is random, then write the SVRG noise as a centred sample.

Define $\Delta_i := \nabla f_i(x_k) - \nabla f_i(\tilde{x})$ and $\bar{\Delta} := \frac{1}{n}\sum_{i=1}^n \Delta_i = \nabla F(x_k) - \nabla F(\tilde{x})$. Then $\xi_{k+1} = \Delta_{i_{k+1}} - \bar{\Delta}$, so $\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = 0$ and

$$\mathbb{E}\big[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k\big] \le \mathbb{E}\big[\|\Delta_{i_{k+1}}\|^2 \mid \mathcal{F}_k\big].$$

Using L-Lipschitz continuity of each ∇f_i gives $\|\Delta_{i_{k+1}}\| \le L\|x_k - \tilde{x}\|$, hence

$$\mathbb{E}\big[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k\big] \le L^2\|x_k - \tilde{x}\|^2.$$

Finally, $\|x_k - \tilde{x}\|^2 \le 2\|x_k - x^*\|^2 + 2\|\tilde{x} - x^*\|^2$ yields (3.3.2) with $C = 2L^2$. □

Remark. This scaling property strictly contrasts with classical SGD, where the sampling noise variance generally remains bounded away from zero even at the exact minimiser x^*.

From the SA viewpoint, the drift term $h(x) = \nabla F(x)$ is unchanged, so the stability from strong convexity is the same as in Chapter 2. What changes is the noise. Instead of being bounded with a non-zero variance floor at the solution, it is iterate-dependent and shrinks as x_k and $\tilde{x}$ approach x^*. This is the reason variance reduction can escape the classical noise limitations.

Corollary 3.3.3 (Constant Step Size Convergence)**.** *Because the stochastic gradient noise variance strictly vanishes near the minimiser x^*, the classical Robbins–Monro summability condition $\sum_{k=0}^\infty \alpha_k^2 < \infty$ is not required to suppress residual variance. Consequently, variance-reduced algorithms permit the use of a constant step size $\alpha_k = \alpha > 0$ while achieving exact convergence under strong convexity.*

Proof. The role of $\sum_k \alpha_k^2 < \infty$ in the classical Robbins–Monro proof is purely to make the accumulated additive noise finite. In the classical case,

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \le (1 - c\alpha_k)V_k + C\alpha_k^2, \tag{3.3.3}$$

with $C > 0$ independent of k. If we take $\alpha_k \equiv \alpha > 0$, then iterating and taking total expectations gives

$$\mathbb{E}[V_k] \;\le\; (1-c\alpha)^k V_0 \;+\; C\alpha^2 \sum_{t=0}^{k-1} (1-c\alpha)^t \;\le\; (1-c\alpha)^k V_0 \;+\; \frac{C}{c}\alpha,$$

so the error cannot converge to 0. This is exactly the Robbins–Monro issue: to make the cumulative noise contribution disappear, we must enforce $\alpha_k \to 0$ and $\sum_k \alpha_k^2 < \infty$.

However, for variance-reduced methods the additive term scales with a quantity that also decays as the iterates converge. We obtain a recursion of the form

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \le (1-c\alpha)V_k + \alpha^2\,\eta_k, \qquad \text{with } \eta_k \to 0, \tag{3.3.4}$$

for some fixed $\alpha > 0$ small enough that $1 - c\alpha \in (0,1)$. Iterating (3.3.4) gives

$$\mathbb{E}[V_k] \le (1-c\alpha)^k V_0 + \alpha^2 \sum_{t=0}^{k-1} (1-c\alpha)^{k-1-t}\,\mathbb{E}[\eta_t].$$

Because the geometric weights $(1-c\alpha)^{k-1-t}$ decay rapidly, the sum puts very little weight on past noise terms. Since $\eta_t \to 0$, the remaining contributions become arbitrarily small as $k \to \infty$, so there is no fixed noise floor. Therefore the condition $\sum_k \alpha_k^2 < \infty$ is no longer needed purely to eliminate an error plateau. □

Theorem 3.3.4 (Linear Convergence of SVRG Epochs [6])**.** *Let F be μ-strongly convex and L-smooth. With a sufficiently small constant step size α and epoch length m, the sequence of snapshot iterates $\tilde{x}^{(s)}$ generated by SVRG satisfies:*

$$\mathbb{E}[F(\tilde{x}^{(s)}) - F(x^*)] \le \rho^s \big(F(\tilde{x}^{(0)}) - F(x^*)\big) \tag{3.3.5}$$

for some contraction factor $\rho \in (0,1)$.

Proof Sketch. While the full proof is lengthy, the mechanism follows directly from the stochastic approximation properties established in Lemma 3.3.2. The proof can be broken into three steps:

1. **Single-Step Descent:** Using the L-smoothness of F, taking a step with the unbiased SVRG estimator g_{k+1} gives an expected single-step decrease, penalised by the variance of the estimator $\mathbb{E}[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k]$.

2. **Variance Substitution:** We substitute the vanishing variance bound derived in Lemma 3.3.2, which bounds the noise strictly in terms of $\|x_k - x^*\|^2$ and $\|\tilde{x} - x^*\|^2$.

3. **Telescoping:** Summing the single-step inequalities over m inner iterations of the epoch allows the negative drift terms (from strong convexity) to dominate the accumulated variance terms. Choosing $\alpha < \frac{1}{4L}$ and m sufficiently large ensures the ratio of the final error to the initial error of the epoch is strictly bounded by $\rho < 1$, yielding linear convergence.

□

Extending to Prox-SVRG: For composite objectives $P(x) = F(x) + R(x)$, Prox-SVRG replaces the gradient step with a proximal step:

$$x_{k+1} = \mathrm{prox}_{\alpha R}\big(x_k - \alpha\, g_{k+1}\big).$$

Using the proximal optimality condition, this can be written in implicit SA form: there exists $v_{k+1} \in \partial R(x_{k+1})$ such that

$$x_{k+1} = x_k - \alpha\big(\nabla F(x_k) + v_{k+1} + \xi_{k+1}\big), \qquad \xi_{k+1} := g_{k+1} - \nabla F(x_k).$$

Hence, the proximal term changes the drift to the composite first-order condition, while the same variance-reduction mechanism controls the sampling noise. In particular, Prox-SVRG inherits the same vanishing-noise behaviour expressed in (3.3.2).

3.4 Noise Behaviour and Convergence

Noise at the Optimum Chapter 2 showed that classical SGD-type schemes are Robbins–Monro recursions in which the noise remains active at the minimiser. By contrast, SVRG-type estimators exploit the finite-sum structure to produce a noise term whose conditional variance decays as the iterates approach the solution. This subsection analyses the difference in form.

Persistent noise in classical SGD: In the ERM setting, the SGD noise term $\xi_{k+1}^{\mathrm{SGD}} = \nabla f_{i_{k+1}}(x_k) - \nabla F(x_k)$ is a martingale difference (Section 2.2) with bounded conditional second moment, but it does not vanish at the minimiser in general (Section 2.2). Unless all component gradients agree at x^*, we have

$$\mathbb{E}\big[\|\xi_{k+1}^{\mathrm{SGD}}\|^2 \mid \mathcal{F}_k\big]\Big|_{x_k = x^*} > 0,$$

so the noise has a non-zero variance floor at the solution.

Vanishing noise in SVRG/Prox-SVRG: Within an epoch, SVRG defines the corrected estimator $g_{k+1} = \nabla f_{i_{k+1}}(x_k) - \nabla f_{i_{k+1}}(\tilde{x}) + \nabla F(\tilde{x})$ and the associated SA noise $\xi_{k+1} := g_{k+1} - \nabla F(x_k)$ (3.3.1). If $x_k = \tilde{x} = x^*$ then $g_{k+1} = \nabla F(x^*)$ and hence $\xi_{k+1} = 0$ a.s.

Generally, under L-Lipschitz continuity of each ∇f_i, the conditional variance can be bounded in terms of the distances to the minimiser:

$$\mathbb{E}\big[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k\big] \;\le\; C\big(\|x_k - x^*\|^2 + \|\tilde{x} - x^*\|^2\big), \tag{3.3.2}$$

so $\mathbb{E}[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k] \to 0$ as $x_k, \tilde{x} \to x^*$. (See proof of Lemma 3.3.2.)

The same conclusion applies to Prox-SVRG. The proximal step changes how R is handled, but it does not reintroduce sampling noise. The variance reduction mechanism is entirely contained in g_{k+1}.

Constant Step Sizes and Linear Convergence This section explains why the vanishing noise removes the Robbins–Monro noise-floor issue and why this allows for constant step size convergence under strong convexity.

Recall in the classical setting (Chapter 2), we obtain a bound of the form

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \;\leq\; (1 - c\alpha_k)V_k + C\alpha_k^2,$$

where $V_k = \|x_k - x^*\|^2$ and the additive term $C\alpha_k^2$ comes from a non-vanishing conditional variance at x^*. With a constant step size $\alpha_k \equiv \alpha$, this produces an $O(\alpha)$ error plateau rather than exact convergence, which is why Robbins–Monro imposes $\sum_k \alpha_k^2 < \infty$.

For SVRG/Prox-SVRG, the variance bound (3.3.2) implies that the additive term is no longer a constant multiple of α^2 near the solution, it is multiplied by a quantity that tends to zero as $(x_k, \tilde{x}) \to (x^*, x^*)$. Consequently, with α constant, the recursion behaves like

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \;\leq\; (1 - c\alpha)V_k \;+\; \alpha^2\eta_k, \qquad \eta_k \to 0,$$

so as a result, there is no fixed noise floor that forces a step-size decay purely for noise control. This means that the Robbins–Monro condition $\sum_k \alpha_k^2 < \infty$ is no longer needed once the noise vanishes near the minimiser (see Theorem 3.3.3).

When strong convexity ensures contraction and there is no constant noise floor, the error recursion is essentially a contraction. In this case, for a constant step size $\alpha > 0$ chosen small enough, the expected error decays geometrically:

$$\mathbb{E}[V_k] \leq (1 - c\alpha)^k V_0 \;+\; (\text{a term that vanishes as } k \to \infty).$$

Hence variance reduction improves rates not by changing the drift (the mean field), but by changing the noise so that it dies out near x^*.

3.5 Summary

In stochastic approximation terms, SVRG and Prox-SVRG keep the same drift as SGD (the full gradient, or the composite first-order condition), but replace persistent sampling noise with noise whose conditional variance vanishes as $(x_k, \tilde{x}) \to (x^*, x^*)$. The key bound (3.3.2) makes this precise.

This removes the classical Robbins–Monro noise-floor issue. Constant step sizes no longer necessarily lead to a non-zero error due to persistent noise, and under strong convexity the dynamics become contraction-dominated, leading to geometric decay once the iterates enter a neighbourhood of the minimiser where the noise is small.

Chapter 4 tests these predictions in a controlled finite-sum quadratic problem by comparing noise behaviour near x^* and the resulting convergence profiles for SGD/Prox-SGD versus SVRG/Prox-SVRG under constant and diminishing step sizes.

Chapter 4

Experiments

This chapter empirically tests the stochastic-approximation (SA) interpretation developed in Chapters 2–3. The theoretical claim is that classical SGD-type methods are Robbins–Monro recursions with persistent sampling noise at the minimiser, which forces diminishing step sizes and yields sublinear convergence. By contrast, variance-reduced methods such as SVRG and Prox-SVRG preserve the same mean field but modify the noise process so that its conditional variance vanishes as the iterates approach the solution, enabling constant step sizes and linear convergence under strong convexity.

4.1 Experimental Design

Problem Setup To isolate the noise mechanism discussed in Chapters 2–3 in a setting where the global optimum is well-defined and computable, we use a synthetic least-squares model with a finite-sum structure. This allows for a controlled, strongly convex problem in which we can (i) compute accurate reference optima and (ii) directly measure the variance of the stochastic gradient estimators.

Data generation: Let n denote the number of samples and d the dimension. We generate a design matrix $A \in \mathbb{R}^{n\times d}$ with i.i.d. Gaussian entries,

$$A_{ij} \sim \mathcal{N}(0,1),$$

a data generating vector

$$x_{\text{true}} \sim \mathcal{N}(0, I_d),$$

and responses

$$b = Ax_{\text{true}} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n),$$

independent of (A, x_{true}). This construction ensures that the objective is a noisy but well-specified quadratic model.

Standardisation: Following common practice in least-squares modelling, we standardise the dataset to improve numerical stability and interpretability. Specifically, each column of A is

centred and scaled to have mean 0 and standard deviation 1, and b is centred to have mean 0. This step reduces scaling effects across coordinates and improves the conditioning of $A^T A$, thereby stabilising the Lipschitz constant of the gradient.

Finite-sum structure: Define per-sample losses

$$f_i(x) := \frac{1}{2}\big(a_i^\top x - b_i\big)^2 + \frac{\mu}{2}\|x\|^2, \qquad i = 1, \dots, n,$$

where $a_i^\top$ is the i-th row of A. The smooth objective is then the empirical average

$$F(x) := \frac{1}{n}\sum_{i=1}^{n} f_i(x) = \frac{1}{2n}\|Ax - b\|^2 + \frac{\mu}{2}\|x\|^2,$$

which matches the stochastic approximation / ERM framework of Chapter 2.

Smooth objective (ridge least squares)

We consider the strongly convex ridge least-squares objective

$$F(x) = \frac{1}{2n}\|Ax - b\|^2 + \frac{\mu}{2}\|x\|^2.$$

The ridge term ensures μ-strong convexity, so F has a unique minimiser x_F^*. Since F is quadratic, the minimiser can be computed exactly by solving the equations:

$$\left(\frac{1}{n}A^\top A + \mu I\right) x_F^* = \frac{1}{n}A^\top b.$$

We denote the optimal value by $F^* := F(x_F^*)$. This closed-form reference solution provides a reliable benchmark for objective gaps and stationarity measures.

Composite objective (ridge + ℓ_1)

To test the proximal theory in Chapters 2–3, we also consider a composite objective of the form

$$P(x) = F(x) + \lambda\|x\|_1 = \frac{1}{2n}\|Ax - b\|^2 + \frac{\mu}{2}\|x\|^2 + \lambda\|x\|_1,$$

which is μ-strongly convex because of the ridge term $\frac{\mu}{2}\|x\|^2$. The ℓ_1 penalty encourages sparsity and makes the objective non-smooth, so the correct first-order stationarity concept is expressed via the proximal gradient mapping rather than ∇P.

Unlike the smooth case, there is no closed-form expression for the minimiser x_P^*. We therefore compute a high-accuracy reference solution using ISTA (proximal gradient descent) with step size $\alpha \approx 1/L$, iterating until the gradient-mapping residual falls below a tight tolerance. We denote the corresponding optimal value by $P^* := P(x_P^*)$.

Why this problem works: This experimental design was chosen to match the theory of this report:

- It is a *finite-sum* objective, so SGD and SVRG-type estimators are well-defined.
- The ridge term enforces strong convexity, ensuring the SA drift stabilises. This is the setting where variance reduction yields linear convergence
- Exact or high-accuracy reference solutions (F^*, x_F^*) and (P^*, x_P^*) are available, allowing objective gaps and stationarity to be evaluated meaningfully.

Fixed experimental parameters (reproducibility): Unless stated otherwise, all experiments in this chapter use the fixed configuration $n = 2000$, $d = 50$, observation noise $\sigma = 0.1$, ridge parameter $\mu = 10^{-4}$, and ℓ_1 regularisation parameter $\lambda = 2 \times 10^{-2}$, with initial iterate $x_0 = 0$. The data are generated with a fixed pseudorandom seed (seed = 42 for the dataset), and each algorithm is run over multiple independent sampling seeds (here: $\{0, 1, 2, 3, 4\}$) with results reported as mean curves with a ± 1 standard deviation band.

Algorithms Compared

Setting	Method	Description
Smooth	SGD-const	$x_{k+1} = x_k - \alpha \nabla f_{i_{k+1}}(x_k)$ (constant α)
Smooth	SGD-RM	Same update with Robbins–Monro schedule $\alpha_k = \frac{a}{k+b}$
Smooth	SVRG	Epoch-based updates with estimator $g = \nabla f_i(x) - \nabla f_i(\tilde{x}) + \nabla F(\tilde{x})$ and constant step size
Composite	Prox-SGD-const	$x_{k+1} = \mathrm{prox}_{\alpha\lambda\|\cdot\|_1}\big(x_k - \alpha \nabla f_{i_{k+1}}(x_k)\big)$
Composite	Prox-SGD-RM	Same proximal update with $\alpha_k = \frac{a}{k+b}$
Composite	Prox-SVRG	Proximal update with SVRG estimator and constant step size

Table 4.1: Shows the algorithms used in the controlled benchmark experiments.

Step sizes, conditioning, and budget We estimate the smoothness and curvature parameters from the quadratic model. For the ridge least-squares objective, the Hessian is

$$H = \frac{1}{n} A^\top A + \mu I.$$

We take L as the largest eigenvalue of H and m as the smallest eigenvalue, so the condition number is $\kappa = L/m$. In our configuration, κ is close to 1, meaning the problem is well-conditioned. This is useful because it keeps conditioning effects small, so the main differences we observe are driven by the noise (and how variance reduction changes it).

For a fair runtime comparison, we measure cost by the number of component-gradient evaluations (calls to ∇f_i). One SGD step uses one component gradient. One SVRG epoch costs n component gradients to compute the full gradient $\nabla F(\tilde{x})$, plus $2m_{\text{in}}$ component-gradient calls in the inner loop (one for $\nabla f_i(x)$ and one for $\nabla f_i(\tilde{x})$ per inner iteration). We therefore set the SGD run length so that SGD and SVRG use the same total number of component-gradient calls. Thus the per-epoch SVRG cost is $n + 2m_{\text{in}}$, and the total budget is epochs $\times$ $(n + 2m_{\text{in}})$ component-gradient calls; SGD is run for this many iterations.

Evaluation metrics We log three metrics as functions of component-gradient calls:

1. **Objective gap:** $F(x) - F^*$ (smooth) or $P(x) - P^*$ (composite).
2. **Estimator variance diagnostic:** $\mathbb{E}\|g_i(x) - \nabla F(x)\|^2$, estimated by subsampling indices at each log point.
3. **Stationarity:** $\|\nabla F(x)\|$ (smooth) and the composite gradient mapping norm $\|G_\alpha(x)\|$ (composite), where

$$G_\alpha(x) = \frac{1}{\alpha}\left(x - \mathrm{prox}_{\alpha\lambda\|\cdot\|_1}(x - \alpha\nabla F(x))\right).$$

4.2 Experimental Results

Mechanism plots: smooth case Figure 4.1 reports objective gap, estimator variance, and stationarity for SGD-const, SGD-RM, and SVRG.

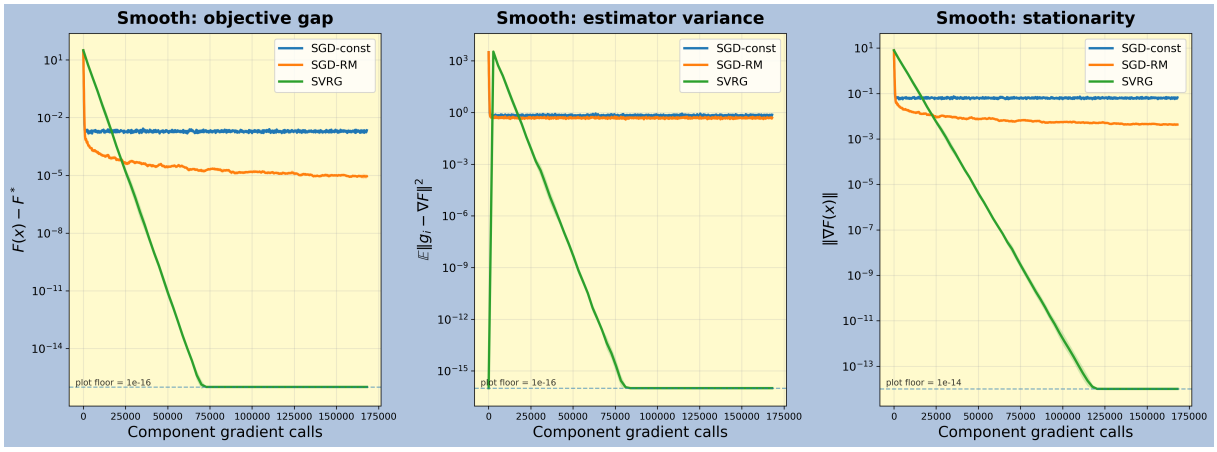

Figure 4.1: Shows the mechanism plots for the smooth problem: objective gap $F(x) - F^*$ (left), estimator variance $\mathbb{E}\|g_i - \nabla F\|^2$ (middle), and stationarity $\|\nabla F(x)\|$ (right), plotted against component-gradient calls. Dashed lines indicate metric-specific plotting floors (double-precision)

Objective gap (left panel): We can see that initially, all methods reduce $F(x) - F^*$ rapidly from the initial iterate. This is because strong convexity pulls the iterates towards x^*. We then see the curves separate according to whether their noise remains persistent or vanishes near the solution. With constant step size, SGD shows a clear error floor (around 10^{-3} for this experiment). After the initial drop the curve essentially becomes flat except for small fluctuations. Because of the non-zero noise, Robbins–Monro with constant step size cannot converge exactly, instead settling into an $O(\alpha)$ neighbourhood of the minimiser. Under the Robbins–Monro step schedule (SGD-RM), the objective gap decreases well below that constant-step floor. It decreases at a slower, sub-linear rate. In contrast to these, SVRG shows an approximately linear decay on the log scale over a long range and reaches the plotting floor well within the gradient call budget. This is consistent with the theory.

Estimator variance (middle panel): The middle panel explains why the objective-gap curves separate. For both SGD-const and SGD-RM, the variance estimate stays roughly flat at an $O(1)$

level. This shows that using a Robbins–Monro step-size schedule does not reduce the variance of the stochastic gradient estimator itself, it only reduces how strongly that variance affects the updates because the steps get smaller. SVRG behaves differently. Its variance drops drastically and eventually hits the plotting floor. This is exactly what the control-variate mechanism is supposed to do. The SVRG noise can be bounded by a constant times $\|x_k - x^*\|^2 + \|\tilde{x} - x^*\|^2$, so as both the current iterate and the snapshot approach x^*, the estimator variance shrinks toward zero. This results in the fixed noise floor present in classical SGD disappearing (which is why SVRG can keep making progress with a constant step size).

Stationarity (right panel). The stationarity panel shows the same split as the objective gap. With a constant step size, SGD levels off at a non-zero gradient norm (around 10^{-1}), meaning it does not settle at a true stationary point, but keeps bouncing around near x^*. With the Robbins–Monro schedule, the stationarity keeps decreasing (down to about 10^{-3} by the end), but the improvement is slow because the steps shrink. With SVRG, the curve is close to a straight line on the log scale until it hits the plotting floor. What stands out is that SVRG drives the objective gap, variance diagnostic, and stationarity down at the same time. This aligns with the SA viewpoint. The drift is similar across all methods, but variance reduction changes the noise so it dies out near x^*, which lets a constant step size keep contracting toward the solution.

Mechanism plots: composite case Figure 4.2 repeats the same analysis for the composite objective $P = F + \lambda\|x\|_1$, using the gradient-mapping norm $\|G_\alpha(x)\|$ as the stationarity metric.

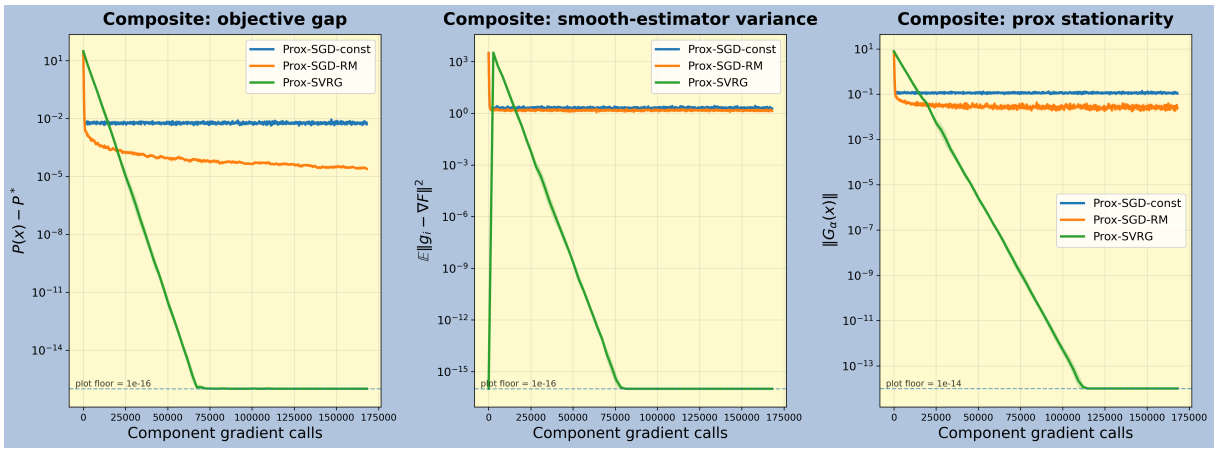

Figure 4.2: Shows the mechanism plots for the composite problem: objective gap $P(x) - P^*$ (left), smooth-estimator variance $\mathbb{E}\|g_i - \nabla F\|^2$ (middle), and prox stationarity $\|G_\alpha(x)\|$ (right), plotted against component-gradient calls. Dashed lines indicate metric-specific plotting floors (double-precision)

Overall, the composite results mirror the smooth case. After an initial decrease, the methods separate according to their noise behaviour near the solution. Prox-SGD with a constant step size plateaus in both objective gap and prox-stationarity, indicating a persistent noise floor. Prox-SGD with Robbins–Monro steps continues to decrease but with a slow tail, since diminishing step sizes reduce the effect of noise at the cost of slower progress. By contrast, Prox-SVRG drives the objective gap and stationarity down to the plotting floor, and the middle panel shows that

this is because the variance of the variance-reduced estimator collapses drastically, whereas the plain stochastic gradient variance remains $O(1)$.

Summary statistics

Case	Method	Rate (slope / p)	R^2	Final gap (mean ± std)	Var. at end (mean ± std)
	SGD-const	-5×10^{-5}	0.00	$1.98 \times 10^{-3} \pm 1.2 \times 10^{-5}$	$7.45 \times 10^{-1} \pm 5.0 \times 10^{-2}$
	SGD-RM	-0.954	0.96	$9.97 \times 10^{-6} \pm 9.1 \times 10^{-7}$	$5.00 \times 10^{-1} \pm 7.0 \times 10^{-2}$
	SVRG	-6×10^{-4}	1.00	$\leq 10^{-16}$ (floor)	$\leq 10^{-16}$ (floor)
	Prox-SGD-const	$+4 \times 10^{-6}$	0.00	$6.06 \times 10^{-3} \pm 1.4 \times 10^{-4}$	$1.98 \times 10^{0} \pm 5.4 \times 10^{-1}$
	Prox-SGD-RM	-0.948	0.95	$3.21 \times 10^{-5} \pm 4.0 \times 10^{-6}$	$1.39 \times 10^{0} \pm 2.9 \times 10^{-1}$
	Prox-SVRG	-6×10^{-4}	1.00	$\leq 10^{-16}$ (floor)	$\leq 10^{-16}$ (floor)

Table 4.2: Shows the summary statistics for the algorithms considered

Table 4.2 aggregates quantitative summaries across five independent seeds for both the smooth and composite experiments. For each method, we report:

- **Rate**: how fast the objective gap decreases in the later stage of the run. For constant-step and SVRG-type methods, it is the slope of log(gap) versus component-gradient calls (geometric decay). For Robbins–Monro step sizes, it is the fitted power-law exponent p in gap $\sim$ calls^p (sublinear decay).
- R^2: how well the chosen model (linear on a log scale or power law) fits the observed decay.
- **Final gap**: mean ± standard deviation of the objective gap over the final 25% of the run.
- **Variance at end**: mean ± standard deviation of $\mathbb{E}\|\nabla f_i(x_k) - \nabla F(x_k)\|^2$ at the final logged iterate.

All rates are measured per component-gradient call. Consequently, they are not directly comparable to iteration-based theoretical rates without rescaling by the cost per epoch.

Constant step-size methods. For SGD-const and Prox-SGD-const, the fitted slopes are numerically close to zero and $R^2 \approx 0$. This matches what we see in the plots. After an initial drop, the objective gap levels off. The final variance remains large, showing that the noise does not disappear. This is exactly what stochastic approximation predicts. Constant step size and persistent noise results in a noise floor rather than full convergence.

Robbins–Monro step sizes. For SGD-RM and Prox-SGD-RM, the fitted exponents $p \approx -0.95$ indicate decay close to $\mathcal{O}(1/k)$, as expected. The high R^2 values (approximately 0.95–0.96) show that the power-law model fits the data well. Compared to constant-step methods, the final gaps are noticeably smaller, reflecting the fact that shrinking step sizes progressively dampen the effect of noise.

Variance-reduced methods. SVRG and Prox-SVRG show clear linear decay with $R^2 \approx 1.00$. In both smooth and composite settings, the objective gap and estimator variance drop to the numerical floor (approximately 10^{-16}). This indicates that the stochastic gradient variance effectively vanishes near the minimiser. Unlike classical SGD, there is no persistent noise floor. The control-variate mechanism causes the variance to shrink with the distance to the snapshot, allowing constant step sizes to keep contracting toward the solution.

Overall interpretation: Persistent noise at the optimum creates a steady-state error when the step size is kept constant, so the method eventually stalls. Robbins–Monro step-size schedules reduce this error, but only at a sublinear rate, because forcing $\sum \alpha_k^2 < \infty$ gradually suppresses the cumulative noise. Variance reduction changes the picture more fundamentally. By altering the structure of the noise itself, it allows constant step sizes to achieve linear convergence. The close agreement between the fitted rates, final objective gaps, and estimator variance in both the smooth and composite cases reinforces the main point of this report: these methods do not improve performance by changing the drift, but by changing what happens to the noise asymptotically.

Extension: Ill-Conditioned Comparison As an extension to the main experiment, we also test these algorithms in an ill-conditioned setting. The goal is to see whether the conclusions drawn in the well-conditioned case remain valid when the optimisation landscape becomes significantly more challenging.

We construct a finite-sum ridge least-squares problem with $n = 2000$ samples and $d = 50$ dimensions, targeting a condition number of order 10^4 (computed $\kappa \approx 5 \times 10^3$). The smooth case uses a small ridge parameter $\mu = 10^{-4}$, while the composite case adds an ℓ_1 regularisation term with $\lambda = 5 \times 10^{-4}$. Each method is run with five random seeds and a fixed computational budget of 1.2×10^6 component-gradient calls. For SVRG-type methods, this corresponds to 120 epochs with 4000 inner iterations per epoch. Final-stage statistics are computed over the last 25% of iterates.

This setup allows us to compare constant step-size methods, Robbins–Monro schedules, and variance-reduced schemes under a much larger condition number, where slow directions in the curvature should amplify any limitations in the noise model.

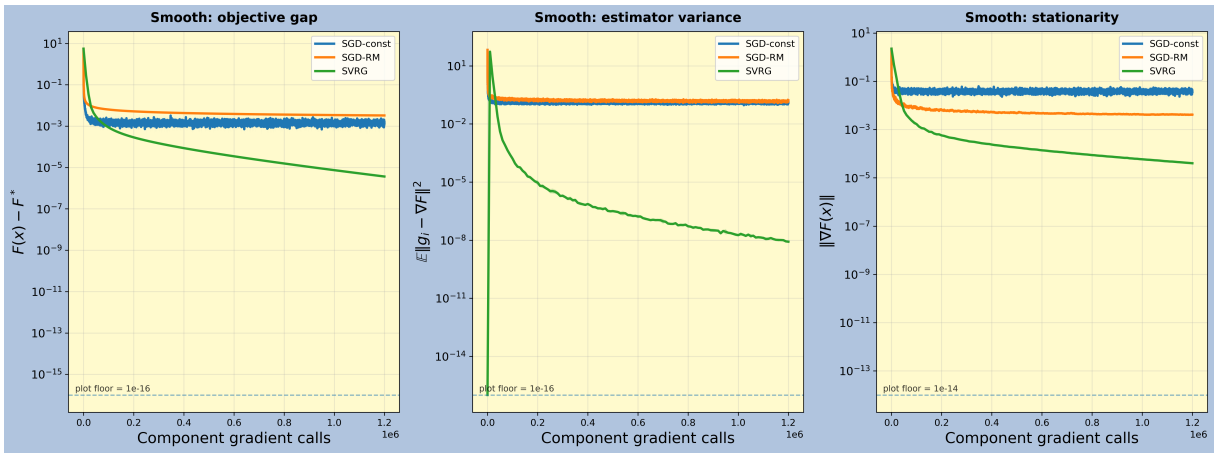

Figure 4.3: Shows the mechanism plots for the ill-conditioned smooth problem: objective gap $F(x) - F^*$ (left), estimator variance $\mathbb{E}\|g_i - \nabla F\|^2$ (middle), and stationarity $\|\nabla F(x)\|$ (right), plotted against component-gradient calls. Dashed lines indicate metric-specific plotting floors (double-precision)

Comparing the well-conditioned and ill-conditioned smooth experiments makes the effect of the condition number much more visible. In the well-conditioned case ($\kappa \approx 1.8$), SVRG exhibits clean linear decay and reaches the numerical floor quickly. Under ill-conditioning ($\kappa \approx 5 \times 10^3$), geometric decay persists, but the slope is noticeably flatter and the method no longer reaches machine precision within the same budget.

For the classical methods, we see something quite interesting. The constant step-size scheme seems to bounce around within a noise-induced neighbourhood of the minimiser. The Robbins–Monro schedule still exhibits sublinear decay, but convergence becomes significantly slower, and within the fixed budget it actually performs worse than the constant step-size method. In an ill-conditioned problem, progress along the smallest-eigenvalue directions is already weak; shrinking the step size further reduces movement in those directions, so the method becomes overly cautious and fails to make sufficient progress within the available iterations.

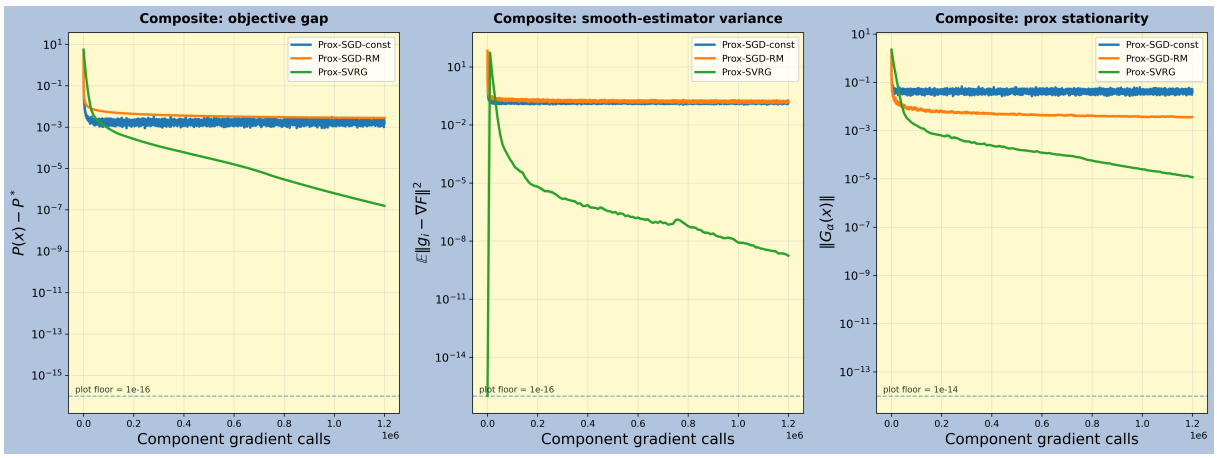

Figure 4.4: Shows the mechanism plots for the ill-conditioned composite problem: objective gap $P(x) - P^*$ (left), smooth-estimator variance $\mathbb{E}\|g_i - \nabla F\|^2$ (middle), and prox stationarity $\|G_\alpha(x)\|$ (right), plotted against component-gradient calls. Dashed lines indicate metric-specific plotting floors (double-precision)

In the composite setting the same behaviour appears as in the smooth case, with the proximal step mainly shifting the scales rather than changing the overall story. Prox-SVRG still shows geometric decay and its estimator variance collapses as the iterates approach the minimiser, whereas the two Prox-SGD variants remain dominated by persistent noise. Again, Prox-SGD-const quickly reaches a noise-limited neighbourhood and then fluctuates within it. Although Prox-SGD-RM continues to decrease sublinearly, the step sizes become too small to make meaningful progress along the ill-conditioned directions within the available budget, so its objective gap remains above the constant-step method by the end. The stationarity curves tell a slightly different story. Robbins–Monro reduces stationarity more, but for objective gap the finite-budget ordering matches the smooth ill-conditioned experiment.

Case	Method	Rate (slope / p)	R^2	Final gap (mean ± std)	Var. at end (mean ± std)
	SGD-const	-0.0000	0.01	$1.55 \times 10^{-3} \pm 5.9 \times 10^{-6}$	$1.33 \times 10^{-1} \pm 1.4 \times 10^{-2}$
	SGD-RM	-0.270	1.00	$3.38 \times 10^{-3} \pm 1.2 \times 10^{-4}$	$1.65 \times 10^{-1} \pm 9.0 \times 10^{-3}$
	SVRG	-0.0000	0.94	$6.55 \times 10^{-6} \pm 2.5 \times 10^{-7}$	$8.29 \times 10^{-9} \pm 7.8 \times 10^{-10}$
	Prox-SGD-const	-0.0000	0.00	$1.83 \times 10^{-3} \pm 8.6 \times 10^{-6}$	$1.56 \times 10^{-1} \pm 1.4 \times 10^{-2}$
	Prox-SGD-RM	-0.277	0.99	$2.84 \times 10^{-3} \pm 9.7 \times 10^{-5}$	$1.76 \times 10^{-1} \pm 9.6 \times 10^{-3}$
	Prox-SVRG	-0.0000	0.99	$5.46 \times 10^{-7} \pm 4.2 \times 10^{-8}$	$1.77 \times 10^{-9} \pm 3.2 \times 10^{-10}$

Table 4.3: Shows the (ill-conditioned) summary statistics for the algorithms considered

Comparing the well-conditioned (Table 4.2) and ill-conditioned (Table 4.3) summary statistics highlights how strongly the condition number affects the classical methods while leaving the behaviour of variance reduction largely unchanged. In the well-conditioned case ($\kappa \approx 1.8$), SGD-RM and Prox-SGD-RM show near-$\mathcal{O}(1/k)$ behaviour, with fitted exponents close to -1, and reach final gaps in the 10^{-6}–10^{-5} range. Under ill-conditioning ($\kappa \approx 5 \times 10^3$), the fitted exponents drop to about -0.27, and the final gaps are roughly two orders of magnitude larger within the same budget (closer to three orders in the smooth case). By contrast, the constant step-size methods plateau in both settings. Their fitted slopes are essentially zero and their estimator variance stays bounded away from zero, consistent with a persistent noise floor.

Variance-reduced methods behave differently. In both experiments, SVRG and Prox-SVRG drive the estimator variance essentially to zero and reduce the objective gap drastically. Ill-conditioning flattens the geometric slope and slows the time scale of contraction, but does not alter the linear convergence behaviour or the vanishing-noise mechanism. Overall, the tables are consistent with the stochastic-approximation view. Conditioning controls how quickly contraction occurs, while variance reduction determines if the noise persists or vanishes near the solution.

4.3 Discussion and Interpretation

Across both the smooth and composite objectives, the experiments support the same SA message from Chapters 2–3. The main difference between classical and variance-reduced methods is not the drift (mean field), but the noise behaviour near the solution.

For SGD and Prox-SGD, the gradient estimator does not vanish at x^*. With a constant step size, this leads to a clear error floor. After an initial decrease, the objective gap and stationarity

flatten out. The Robbins–Monro step-size schedule improves accuracy by shrinking α_k, but only at a sublinear rate. Importantly, the variance itself does not decrease, its effect is merely damped.

On the other hand, SVRG and Prox-SVRG keep the same drift $h(x) = \nabla F(x)$ (or the composite first-order condition in the proximal case), but replace the sampling noise with an iterate-dependent term whose conditional second moment shrinks as $(x_k, \tilde{x}) \to (x^*, x^*)$. This is visible in the variance diagnostics, which drop by many orders of magnitude. Once the noise vanishes, the contraction from strong convexity takes over, producing a near-linear decay observed in the plots. This empirical collapse of variance perfectly mirrors the original theoretical guarantees of SVRG established by Johnson and Zhang [6]. Furthermore, the parallel separation observed in the composite setting, where Prox-SGD plateaus but Prox-SVRG maintains geometric convergence, aligns precisely with Xiao and Zhang's findings on elastic net regularisation, confirming that while the proximal step resolves non-smoothness, it is the variance reduction that controls the asymptotic rate [5].

The ill-conditioned extension mirrors this structure but stretches the time scale. The increase in κ causes contraction along slow directions to weaken. For Robbins–Monro step-size schedules, the fitted exponents drop sharply and final gaps are much larger within the same budget. In fact, in both smooth and composite cases, the constant step-size method ends with a smaller objective gap than the diminishing step-size variant. This finite-budget failure is well-documented in the non-asymptotic analysis of Moulines and Bach [9], when the step-size decay is too aggressive relative to the strong convexity parameter (the smallest eigenvalue), the theoretical constants in the $\mathcal{O}(1/k)$ rate expand drastically. The steps shrink faster than the weak curvature directions can converge, causing the method to stall prematurely. The constant step methods, while still plateauing at a variance floor, avoid this artificial stalling.

Variance-reduced methods remain different. Ill-conditioning flattens their geometric slope, but the variance still collapses and contraction ultimately dominates. As highlighted by Gower et al.'s unified framework for variance-reduced methods [10], the linear convergence rate factor is strictly bound to the condition number κ. Increasing κ pushes the contraction factor closer to 1, causing the exact flattening observed in our plots, yet it does not destroy the underlying vanishing-noise mechanism. In short, conditioning dictates how strong the drift is, but variance reduction decides whether the noise persists. When the noise vanishes, linear convergence survives even in ill-conditioned environments.

Chapter 5

Conclusion

The central question posed at the start of this paper — namely, what fundamentally separates classical stochastic gradient methods from modern variance-reduced schemes — can now be answered clearly. The difference is not the drift, but the noise.

This work connects classical stochastic approximation to modern finite-sum optimisation by making the noise model the main object of comparison. Robbins–Monro updates take the form *drift + noise*: a deterministic mean field that determines stability, plus a martingale-difference noise term that determines how step sizes must be chosen. Writing SGD in this form identifies the mean field as $\nabla F(x)$ and the noise as the sampling error $\nabla f_i(x) - \nabla F(x)$. Even under strong convexity, this noise typically has a non-zero variance floor at the minimiser, so constant step sizes lead to an error plateau. Diminishing step sizes are the mechanism that make the accumulated noise effect finite, but this forces slow sublinear tails.

Proximal methods fit the same template for composite objectives $P(x) = F(x) + R(x)$. The proximal step changes how the non-smooth term is handled, but it does not remove the sampling noise, so Prox-SGD inherits the same persistent-noise limitation as SGD.

Variance reduction changes what matters. SVRG and Prox-SVRG keep the same drift (the full gradient, or the composite first-order condition), but replace the sampling noise by a control-variate term whose conditional variance shrinks with $\|x_k - \tilde{x}\|$ and therefore vanishes as both the iterate and snapshot approach x^*. This removes the Robbins–Monro noise-floor issue and explains why constant step sizes can achieve geometric decay under strong convexity. The experiments isolate this mechanism. SGD/Prox-SGD with constant step sizes rapidly reach a non-zero error floor, Robbins–Monro schedules continue improving but slowly, and (Prox-)SVRG drives objective gaps and stationarity measures down to numerical floors while the variance drops by orders of magnitude.

As an extension to the main experiment, we also explored one of the central limitations of the original setup, ill conditioning. By constructing a problem with $\kappa \approx 5 \times 10^3$, we examined how the drift–noise balance changes when curvature becomes highly uneven. The results show that ill-conditioning mainly weakens contraction and stretches the time scale of convergence, especially for Robbins–Monro step-size schedules. However, it does not change the underlying mechanism identified in this work. Classical methods remain constrained by a persistent variance floor, while variance-reduced methods continue to drive the noise toward zero and retain geometric decay, with a flatter slope.

These experiments suggest that the stochastic-approximation perspective provides a coherent way to understand finite-sum methods beyond the classical setting. That being said, several important directions remain open. This study focuses on quadratic models with bounded variance, extending the analysis to non-quadratic losses, convex objectives that lack strong convexity, or settings with heavy-tailed noise (a problem that's increasingly observed in modern machine learning tasks [11]) would test how robust the drift–noise decomposition remains when standard assumptions weaken.

More broadly, the results indicate that progress in stochastic optimisation is often less about modifying the mean field and more about reshaping the noise. Recent literature modelling stochastic algorithms as continuous-time processes explicitly separates these drift and diffusion components to understand generalisation [12, 13]. Framing algorithms through this lens may provide a useful guide for designing new methods that retain stability while systematically controlling variance in increasingly challenging problems.

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